

TECHNICAL REVIEW DOCUMENT
For
DRAFT RENEWAL / MODIFICATION of OPERATING PERMIT 95OPBO082

CEMEX Construction Materials South, LLC – Lyons Cement Plant
Boulder County
Source ID 0130003

Issued February 1, 2025

The permit was revised January 2025 prior to issuance to make a correction as discussed on page 52. The correction meets the requirements for an administrative amendment in Colorado Regulation No. 3, Part A, Section I.B.1. The administrative amendment procedures in Colorado Regulation No. 3, Part A, Section III do not require public comment or EPA review.

I. Purpose:

This document will establish the basis for decisions made regarding the applicable requirements, emission factors, monitoring plan and compliance status of emission units covered by the renewed Operating Permit proposed for this site. The current Operating Permit was issued March 1, 2017 and last revised November 17, 2017. The expiration date for the permit was March 1, 2022. However, since a timely and complete renewal application was submitted, under Colorado Regulation No. 3, Part C, Section IV.C all of the terms and conditions of the existing permit shall not expire until the renewal Operating Permit is issued and any previously extended permit shield continues in full force and operation.

The source submitted a renewal application on March 1, 2021. Prior to submittal of the renewal application, the source submitted an APEN on May 30, 2019 for an engine that was not identified in the current permit (last revised November 17, 2017). In addition, the source submitted an application on November 15, 2022 to modify the permit to remove the limestone throughput limit for P011 (AIRS pt 011) in Section II, Condition 11.1 of the permit.

This document is designed for reference during the review of the proposed permit by the EPA, the public, and other interested parties. The conclusions made in this report are based on information provided in the renewal application submitted March 1, 2021, the modification application submitted November 15, 2022, additional information submitted on May 30, 2019, May 9, 2023 and October 11, 2023, comments on the draft permit and technical review document received on August 31, 2023, inspection reports and various e-mail correspondence, as well as telephone conversations with the applicant. Please note that copies of the Technical Review Document for the original permit and any Technical Review Documents associated with subsequent modifications of the original Operating Permit may be found in the Division files as well as on the

Division website at www.colorado.gov/cdphe/airTitleV. This narrative is intended only as an adjunct for the reviewer and has no legal standing.

Any revisions made to the underlying construction permits associated with this facility made in conjunction with the processing of this Operating Permit application have been reviewed in accordance with the requirements of Regulation No. 3, Part B, Construction Permits, and have been found to meet all applicable substantive and procedural requirements. This Operating Permit incorporates and shall be considered to be a combined construction/operating permit for any such revision, and the permittee shall be allowed to operate under the revised conditions upon issuance of this Operating Permit without applying for a revision to this permit or for an additional or revised construction permit.

II. Description of Source

This facility manufactures portland cement. Limestone and other raw materials are directed to the crusher, conveyance system, patio or other material stockpiles. These materials are either quarried onsite or delivered to the site via truck or rail. The stockpiled material is typically placed on a belt by means of a front-end loader to be processed through a primary crusher, the dryer, and a secondary crusher. The material from the secondary crusher is stored in raw material storage silos. These storage silos are used for sources of silica, iron, alumina, limestone, and various raw materials. Material from these storage silos is discharged to weigh belts for the formulation of a desired product. The weigh belts discharge to the raw mill. The raw mill mixes and crushes the blended materials and delivers the homogenized material to storage silos. The homogenized material from the storage silos is delivered to the calciner portion of the kiln. Pulverized coal from the coal mill is fired at the bottom of the flash calciner. The calcined material from the calciner then enters the rotary kiln, which is located at a slight incline along its horizontal axis. The material travels towards the clinker discharge end where additional pulverized coal is fired for the clinkering process. The clinker is discharged from the kiln into the clinker cooler where it is cooled by air forced through the clinker bed by undergrate fans. A good percentage of this air is recovered for use as primary air in the kiln combustion process. The cooled clinker is then moved to internal storage in an A-Frame building, or outside storage stockpiles. The stored clinker is the raw material for the finish mill. In the finish mill the clinker is combined with gypsum, ground to a fine material and stored in product silos. The material in the product silos can be loaded for bulk transport, or sent to a packaging system. From an over-all perspective, the manufacturing process may be viewed as two segments -- clinker production and cement production. The clinker storage allows the two processes to operate at different production rates. During periods of low demand for cement, clinker is accumulated. If cement is in high demand, the clinker production can be supplemented by purchase of clinker from other sources. The overall result is the clinker production can operate at a rather steady rate, while the cement production can operate in response to the current or projected demands.

The facility is located near Lyons, 12 miles north of Boulder. The area in which the facility is located is classified as attainment/maintenance for particulate matter less than 10 microns (PM₁₀). Under that classification, all SIP-approved requirements for PM₁₀ will continue to apply in order to prevent backsliding under the provisions of Section 110(l) of the Federal Clean Air Act. This area is classified as nonattainment for ozone and is part of the 8-hr Ozone Control Area as defined in Colorado Regulation No. 7, Part A, Section II.A.1. The 8-hr Ozone Control Area has been classified as a severe non-attainment area effective November 7, 2022.

This facility is categorized as a Non-Attainment Area New Source Review (NANSR) major stationary source (Potential to Emit of VOC and NO_x ≥ 25 tons/year). Future modifications at this facility resulting in a significant net emissions increase (see Regulation No. 3, Part D, Sections II.A.27 and 44) for VOC or NO_x or a modification which is major by itself (Potential to Emit ≥ 25 tons/year of either VOC or NO_x) may result in the application of the NANSR review requirements.

This source is categorized as a Prevention of Significant Deterioration (PSD) major stationary source (Potential to Emit ≥ 100 tons/year) for PM, PM₁₀, SO₂, NO_x and CO. Future modifications at this facility resulting in a significant net emissions increase (see Regulation No. 3, Part D, Sections II.A.27 and 44) or a modification that is major by itself (Potential to Emit ≥ 100 tons/yr) for any pollutant listed in Regulation No. 3, Part D, Section II.A.42 for which the area is in attainment or attainment/maintenance may result in the application of the PSD review requirements.

There are no affected states within 50 miles of the plant. Rocky Mountain National Park, Rawah Wilderness Area and Eagle's Nest Wilderness Area are Federal Class I designated areas within 100 kilometers of the plant.

The summary of emissions that was presented in the Technical Review Document (TRD) for the previous renewal permit issuance (March 1, 2017) has been modified to more appropriately identify the **potential to emit (PTE) and actual emissions** of both criteria and hazardous air pollutants. Emissions (in tons/yr) at the facility are as follows:

Pollutant	Emissions (tons/yr)	
	Potential	Actual
PM	441.11	124.4
PM ₁₀	243.87	69.7
SO ₂	132.90	13.1
NO _x	924.90	288.3
CO	492.90	220.3
VOC	282.80	5.4
Lead (Pb)	6	--
HAPs	11.43	0.7

Actual emissions are from the Division's inventory system, data year 2020.

Potential to emit is summarized by emission unit and/or emission group in the table starting on page 63. The detailed table beginning on page 63 also indicates the basis for the potential emissions.

III. Overview of Modifications

The source submitted applications for minor modifications which were determined to be complete and are now being finalized as part of this renewal.

November 15, 2022 Modification

The purpose of this modification was to remove the limestone throughput limit for P011 (AIRS pt 011) in Section II, Condition 11.1. P011 includes the finish mill (S036) and finish mill auxiliary dust collector (S037). The current permit (last revised November 17, 2017) includes a limit on overall fresh feed to the mill of 631,600 tons/yr, as well as a 15,000 tons/yr limit on limestone. According to the application, the limestone limit is a remnant of a previous modification application.

The modification application qualified as a minor modification because it did not violate any applicable requirements; involve significant changes to monitoring, reporting or recordkeeping requirements; require or change a case-by-case determination of an emission limitation or standard, seek to establish a permit term or condition for which there is no underlying applicable requirement; and is not otherwise required to be processed as a significant modification. In addition, this minor modification did not avoid any requirements that would have applied if this modification were processed as a construction permit modification.

The Division verifies that this modification qualified as a minor modification under the requirements of Colorado Regulation No. 3, Part C, Section X, and did not violate any applicable requirements.

- The modification application does not request any change in allowable emissions and removal of the throughput limit will not affect actual emissions from the affected equipment. Therefore under Section 2.1 of the August 25, 2022 Modeling Guideline Addendum, Section 2.1 since there is no change in emissions, modeling is not required.
- The requirements in Colorado Regulation No. 3, Part B, Section III.D.2 (RACT) do not apply to this modification, because it does not involve a new or modified emissions unit. The modification application does not request an increase in allowable emissions and will not affect actual emissions, thus it does not meet the definition of a modification in Colorado Regulation No. 3, Part A, Section I.B.28 and so the affected emissions units are not considered modified.

- As mentioned in the above bullet, the modification does not request an increase in allowable emissions and will not affect actual emissions, thus the major new source review (NSR) provisions in Colorado Regulation No. 3, Part D were not triggered.

Other Modifications

In addition to the minor modification listed above, this renewal is incorporating changes requested in the renewal application, as well as revisions to update permit language and reflect changes to regulations revised and/or promulgated after the previous permit issuance. These changes are described in other sections of this document.

Other previous modifications that resulted in a revised permit issuance prior to this renewal are listed in Appendix F of the current permit (last revised November 17, 2017) and have been addressed in the permit record documentation associated with those revised permit issuances.

IV. Applicable Requirements

Accidental Release Program (112(r))

Section 112(r) of the Clean Air Act mandates a federal focus on the prevention of chemical accidents. Sources subject to these provisions must develop and implement risk management programs that include hazard assessment, a prevention program, and an emergency response program. They must prepare and implement a Risk Management Plan (RMP) as specified in the Rule.

Based on the information provided by the applicant, this facility is not subject to the provisions of the Accidental Release Prevention Program (Section 112(r) of the Federal Clean Air Act).

Compliance Assurance Monitoring (CAM)

CAM applies to any emission unit that is subject to an emission limitation, uses a control device to achieve compliance with that emission limitation and has potential pre-control emissions greater than major source levels.

CAM applicability was addressed in the TRDs for the both the first (issued March 1, 2008) and second renewal (issued March 1, 2017) permits. The TRD for the second renewal included a detailed discussion of CAM applicability for the emission units included in the summary table in Section I, Condition 4.1 of the permit at that time, thus the analysis will not be included here. Except as described below, no new emission units have been included in the permit since the CAM applicability analysis conducted for the second renewal (issued March 1, 2017) and addressed in the second renewal TRD starting on page 62.

As discussed in the TRD for the second renewal (issued March 1, 2017), the Division revised the CAM requirements for the kiln to rely on the monitoring for PM emissions in the PC MACT, rather than the CAM monitoring that was included in the first renewal (issued March 1, 2008). The TRD for the second renewal (issued March 17, 2017) also indicated that since CAM for the kiln is based on complying with the MACT requirements, no CAM plan was included in the permit. At this time, the Division considers that was not appropriate and is including a CAM plan in the permit for the kiln. The CAM plan is based on the PM monitoring required by the PC MACT.

The evaluation conducted for the second renewal permit only addressed one engine, the A-pit pump, however, additional engines were included in the second renewal permit. None of those engines, as well as the additional engine that will be included in this renewal have control devices, therefore, they are not subject to CAM.

As noted in the current permit (last revised November 17, 2017), the following emission points at this facility use a control device to achieve compliance with an emission limitation or standard to which they are subject and have pre-control emissions that equal or exceed the major source threshold. They are therefore subject to the provisions of the CAM program as set forth in 40 CFR Part 64 as adopted by reference into Colorado Regulation No. 3, Part C, Section XIV:

S001 – Waste Dust Silo; S005 – Raw Materials Dryer; S010 – Raw Material Grinding, S011 – Raw Mill Auxiliary Dust Collector, S012 – Raw Mill Feeders, S016 – Precalciner Kiln; S022 – Kiln Return Dust Silo; S024 - #2 Clinker Silo; S034- #6 Reclaim Feeder; S036 – Finish Mill; S037 – Finish Mill Auxiliary Dust Collector; S043 – Cement Storage Silos A10 and A13; S044 – Cement Storage Silo A7; S045 – Cement Finish Silo A2; S046 – Packhouse West (loading spout); S046 - Packhouse East (loading Spout) - S051 – Top of A Frame Transfer; and S066 – Cement Silo A5

Hazardous Air Pollutant Emissions

Hazardous air pollutants (HAP) are regulated under the National Emission Standards for Hazardous Air Pollutants (NESHAP) promulgated in 40 CFR Parts 61 and 63. The requirements in 40 CFR Part 61 address a specific hazardous air pollutant (HAP), such as benzene, and apply to sources that meet the applicability requirements. Significant changes were made to the Federal Clean Air Act in 1990 with respect to HAP emissions and those changes are reflected in the NESHAPs promulgated under 40 CFR Part 63. The requirements in 40 CFR Part 63 apply to specific source categories, are technology based and are frequently referred to as “Maximum Achievable Control Technology (MACT) requirements”. MACT applicability is also based on whether or not a source is a major source of HAP emissions. A major source of HAPs has the potential to emit 10 tons of any single HAP or 25 tons of combined HAPs.

The initial and revised Title V permit applications only estimated HAP emissions from the kiln and noted hydrochloric acid (HCl) as the significant HAP. The Division would agree that HAP emissions from particulate matter only sources are unlikely to be

significant and that the majority of HAP emissions would be from the kiln, although the dryer would also be a source of HAP emissions.

As indicated in the TRD for the second renewal permit (issued March 1, 2017), the Lyons Cement Plant operated as a major source of HAPs on the first compliance date for the Portland Cement Manufacturing NESHAP (also referred to as the “Portland Cement (PC) MACT”). In the October 23, 1998 Title V permit application, the source used an emission factor for HCl from a September 1996 Portland Cement Association report to EPA. This emission factor (0.0381 lb/ton clinker) indicated that emissions from HCl were above 10 tons/yr thus the facility was a major source for HAPs. A performance test conducted in 2002 indicated that HCl emissions were much lower, slightly less than 4 tons/yr and based on AP-42 emission factors published March 2022, HCl emissions are estimated at over 40 tons/yr. Since there is a high probability that the HCl emissions vary and the source has never taken federally enforceable limits to reflect that HCl emissions may be below the major source level, the Division considers that the source is a major source of HAPs for other NESHAPs issued after the PC MACT.

NESHAPs

Portland Cement Manufacturing Industry (40 CFR Part 63 Subpart LLL)

As indicated in previous TRDs for this permit, the PC MACT applies to this facility and the provisions have been included in the permit since initial permit issuance (February 1, 2000). Significant revisions were made to the PC MACT in 2013 and the TRD for the second renewal (issued March 1, 2017) discusses those changes in detail and how the requirements are addressed in the permit.

The PC MACT applies to both major and minor sources of HAPs. Affected sources (see 63.1340(b)) include each kiln (including alkali bypasses and inline coal mills); clinker cooler; raw mill; finish mill; raw material dryer; raw material, clinker or finished product storage bin; conveying system transfer point (including those associated with coal preparation used to convey coal from the mill to the kiln); bagging and bulk loading and unloading system; and open clinker storage pile.

The following sources at this facility are subject to the PC MACT:

- P002 (raw materials drying)
- P004 (raw material storage silos)
- P005 (raw material grinding)
- P006 (homogenizing and blending)
- P007 (kiln burning)
- P008 (clinker cooling and transfer to storage for finish mill)
- P009 (clinker and gypsum/additive silos and weigh feeders)
- P010 (sheltered (A-Frame) clinker storage and reclamation)

- P015 (outdoor clinker piles and handling)
- P011 & P012 (cement finish mill and auxiliaries)
- P013 (cement silos/packhouse/loading)
- P014 (material handling system – load-in and load-out)
- P007A (handling and processing of CKD and raw material waste dust)
- P050 (cement rail car unloading system)

The Dowe Flats and Lyons Quarry fugitive dust sources are not affected sources as defined in 63.1340(b) and are not subject to PC MACT requirements. This is also the same for haul roads not subject to emission limitations (included in Section II.14 – P018 - general fugitive particulate matter emissions requirements). Section II.14 also includes process fugitives not subject to emission limitations. The description in the table in Section I, Condition 4.1 of the permit says that “process fugitives” include “wind erosion of stock piles and various transfers not vented through a stack (e.g. belt and screw conveyor transfers)”. Stock piles are not affected sources as defined in 63.1340(b) but conveying system transfer points are, therefore, the PC MACT would apply to any conveying system transfer points.

P000 (raw material storage and handling at plant site) is not subject to the PC MACT requirements because there are no affected sources as defined in 63.1340(b) in this group. In addition, the primary (quarry and plant) and secondary crushers, as well as the Dowe Flats quarry conveyor are not subject to the PC MACT requirements because crushers and sources subject to NSPS OOO are exempt (see 63.1340(c)). In addition, neither the lime storage silo or lime weigh hopper are affected sources under 63.1340(b), so they are not subject to PC MACT requirements (lime is not a raw material for making Portland cement but is injected into the kiln to reduce HCl emissions).

Revisions were made to the PC MACT on September 11, 2015, July 25, 2016, June 23 and August 22, 2017, August 3 and July 25, 2018, October 7 and November 19, 2020. The September 11, 2015 revisions made technical corrections and clarifications to the July 27, 2015 revisions to the rule (which are included in the current permit (last revised November 17, 2017)). The July 25, 2016 and June 23 and August 23, 2017 revisions addressed an HCl alternative monitoring method. The July 25, 2018 revisions finalized the residual risk and technology review (RTR) for the PC MACT. No revisions to numerical emission limits were made, the revisions reflect corrections and clarifications of the rule requirements and provisions. The August 3, 2018 revisions were a correction to the July 25, 2018 RTR. The October 7, 2020 amendments are revisions to test methods and performance specifications. The November 19, 2020 revisions address reclassification of major sources as area sources. The appropriate revisions will be made to the permit to reflect these changes and are discussed later in this document.

Reciprocating Internal Combustion Engines (40 CFR Part 63 Subpart ZZZZ)

As indicated in the TRD for the second renewal permit (issued March 1, 2017), the engines at this facility are subject to the requirements in MACT ZZZZ and those requirements have been included in the permit. The permit will be revised as necessary to address any revisions to MACT ZZZZ since the previous permit issuance.

Industrial, Commercial and Institutional Boilers and Process Heaters MACT (40 CFR Part 63 Subpart DDDDD)

As discussed in the TRD for the second renewal (issued March 1, 2017), the provisions in 40 CFR Part 63 Subpart DDDDD apply to boilers and process heaters that are located at major sources of HAPs. As noted in the TRD for the second renewal (issued March 1, 2017), there are no process heaters at this facility. There are also no boilers at this facility, therefore the requirements in Subpart DDDDD do not apply.

Gasoline Dispensing Facilities (40 CFR Part 63 Subpart CCCCCC)

The requirements in 40 CFR Part 63 Subpart CCCCCC, “National Emission Standards for Hazardous Air Pollutants for Source Category: Gasoline Dispensing Facilities” apply to gasoline dispensing facilities (GDF) located at area (minor) sources. There is a 3,000 gallon gasoline storage tank at the facility but since the facility is a major source of HAP emissions, the requirements in 40 CFR Part 63 Subpart CCCCCC do not apply.

New Source Performance Standards (NSPS)

Federal NSPS requirements are codified in 40 CFR Part 60 and are typically adopted into Colorado Regulation No. 6, Part A by reference. There have been no changes in applicability since the second renewal permit, but the applicable requirements are being addressed to identify the requirements that apply to the facility. Note that sources that are subject to a specific subpart under 40 CFR Part 60 are also subject to the general provisions under Subpart A. The NSPS general have also been adopted by reference in Colorado Regulation No. 6, Part A.

NSPS Subpart F – Standards of Performance for Portland Cement Plants

Subpart F is discussed in the TRD for the second renewal (issued March 1, 2017), the following discussion provides a more detailed applicability determination.

The following equipment located at a Portland cement plant that commenced construction or modification after August 17, 1971 are affected facilities subject to the requirements in Subpart F: kiln, clinker cooler, raw mill system, finish mill system, raw mill dryer, raw material storage, clinker storage, finished product storage, conveyor transfer points, bagging and bulk loading and unloading systems (see 60.60(a) and (b)).

The Lyons Cement Plant commenced operation in the 1968-1969 time frame, thus the majority of the equipment is not subject to the requirements in NSPS Subpart F because it does not meet the applicability date in 60.60(b). The following equipment

was modified after August 17, 1971 and is potentially subject to the requirements in NSPS F:

- P017 (Dowe Flats and Lyons quarry – fugitive dust sources)
- P017 (Dowe Flats quarry – point source emissions – primary crusher (AIRS pt 027) and belt conveyor (AIRS pt 026))
- P000 (raw material storage and handling at plant site)
- P002 (raw materials drying)
- P007 (kiln burning)
- P009 (clinker and gypsum/additive silos and weigh feeders)
- P010 (sheltered (A-Frame) clinker storage and reclamation)
- P015 (outdoor clinker piles and handling)
- P011 & P012 (cement finish mill and auxiliaries)
- P013 (cement silos/packhouse/loading)
- P007A (handling and processing of CKD and raw material waste dust)
- P050 (cement rail car unloading system)

The Dowe Flats and Lyons quarry fugitive dust sources are not subject to the requirements in NSPS F because these activities are not affected facilities as identified in 60.60(a). In addition, the activities at P000 (raw material storage and handling at plant site) also are not affected facilities as identified in 60.60(a).

As discussed further below, certain equipment at this facility is also potentially subject to the requirements in 40 CFR Part 60 Subpart OOO, “Standards of Performance for Nonmetallic Mineral Processing Plants”. EPA came out with specific guidance regarding the applicability of Subpart F requirements for portland cement plants with on-site nonmetallic mineral processing plants in a June 12, 1995 applicability determination (control number 9600050), which states the following:

Based on information provided by the Office of Air Quality Planning and Standards, we have determined that the raw mill of Subpart F does not include the quarry crushing system (that includes both primary and secondary crushers) which is always subject to Subpart OOO. The product from the secondary crusher is stored in a raw material storage facility until ready for grinding. The nonmetallic mineral processing plant ends after the secondary crusher conveyor and the raw mill system (Subpart F) begins with the raw material storage facility. The first conveyor transfer point subject to Subpart F is the transfer point associated with the conveyor transferring material from the raw material storage to the grinding operations.

Under the applicability criteria discussed above, the first conveyor transfer point potentially subject to the requirements in Subpart F is P004 (raw material storage silos). Thus the NSPS Subpart F does not apply to the Dowe Flats belt conveyor (AIRS pt

026). Crushers are not an affected facility under 60.60(a), thus the Dowe Flats primary crusher (AIRS pt 027) is also not subject to NSPS Subpart F.

The raw materials dryer (P002) is located prior to raw material storage that feeds the raw mill or grinding operations. Although the dryer precedes the point at which EPA's applicability determination considers the Subpart F requirements apply to Portland cement plants co-located with a nonmetallic mineral processing plant, location of the dryer is not appropriate in assessing applicability with the NSPS F. EPA's applicability determination is intended to distinguish when the NSPS F or OOO requirements apply to equipment that may be applicable to either requirement. Dryers are an affected facility under NSPS F but not NSPS OOO, thus dryers would never be subject to NSPS OOO, so location is irrelevant in determining whether the dryer is subject to the requirements in NSPS F. The raw materials dryer is an affected facility under 60.60(a) and was modified after August 17, 1971, thus NSPS F applies to the raw materials dryer.

In summary the following equipment is subject to the requirements in NSPS F.

- P002 (raw materials drying)
- P007 (kiln burning)
- P009 (clinker and gypsum/additive silos and weigh feeders)
- P010 (sheltered (A-Frame) clinker storage and reclamation)
- P015 (outdoor clinker piles and handling)
- P011 & P012 (cement finish mill and auxiliaries)
- P013 (cement silos/packhouse/loading)
- P007A (handling and processing of CKD and raw material waste dust)
- P050 (cement rail car unloading system)

The equipment that is subject to NSPS F is also subject to the PC MACT requirements. 40 CFR Part 60 Subpart F § 60.62(b) states the following:

If you have an affected source subject to this subpart with a different emissions limit or requirement for the same pollutant under another regulation in title 40 of this chapter, once you are in compliance with the most stringent emissions limit or requirement, you are not subject to the less stringent requirement. Until you are in compliance with the more stringent limit, the less stringent limit continues to apply.

NSPS F include PM limits for the kiln and clinker cooler, however, the PC MACT requirements are more stringent.

The NSPS F opacity limit for all equipment except the kiln and clinker cooler is more stringent than the opacity limit in the PC MACT, therefore, the NSPS F opacity limit is included in the permit. For all other equipment and pollutants the PC MACT requirements are more stringent than the NSPS F requirements. The TRD for the

second renewal permit (issued March 1, 2017) provides a more detailed discussion regarding the NSPS F and PC MACT opacity requirements.

NSPS Subpart Y – Standards of Performance for Coal Preparation and Processing Plants

The provisions of this subpart are applicable to certain equipment located at a coal preparation and processing plant that processes more than 200 tons of coal per day.

Coal preparation and processing plant means any facility (excluding underground mining operations) which prepares coal by one or more of the following processes: breaking, crushing, screening, wet or dry cleaning, and thermal drying.

As discussed on page 3 of the TRD for the first renewal permit (issued March 1, 2008), the source does not meet the definition of a coal preparation and processing plant because it does not conduct any of the above listed processes (breaking, crushing, screening, wet or dry cleaning, and thermal drying). Therefore, no part of the facility is a coal preparation and processing plant and no equipment is subject to NSPS Subpart Y.

NSPS Subpart OOO – Standards of Performance for Nonmetallic Mineral Processing Plants

The provisions of this subpart are applicable to any of the equipment listed below at a nonmetallic mineral processing plant that commenced construction or modification after August 31, 1983 (see 60.670(e)).

A nonmetallic mineral processing plant is defined as any combination of equipment that is used to crush or grind any nonmetallic mineral wherever located, including lime plants, power plants, steel mills, asphalt concrete plants, portland cement plants, or any other facility processing nonmetallic minerals except as provided in § 60.670(b) and (c).

Unless otherwise specified, the following affected facilities are subject to requirements in Subpart OOO: each crusher, grinding mill, screening operation, bucket elevator, belt conveyor, bagging operation, storage bin, enclosed truck or railcar loading station (see 60.670(a)(1)).

As specified in 60.670(b), an affected facility (equipment) that is subject to the provisions of Subpart F or that follows in the plant process any facility subject to the provisions Subpart F is not subject to the provisions of this subpart.

As noted above, under the discussion for Subpart F, only equipment prior to raw materials storage for the raw mill or grinding operations is considered for Subpart OOO applicability. Therefore, the following equipment is prior to raw material storage (P004) and is potentially subject to the requirements.

- P017 (Dowe Flats and Lyons quarry – fugitive dust sources)

- P017 (Dowe Flats quarry – point source emissions – primary crusher (AIRS pt 027) and belt conveyor (AIRS pt 026))
- P000 (raw material storage and handling at plant site)
- P001 (primary crusher (plant site))
- P002 (raw materials drying)
- P003 (secondary crusher)

The Dowe Flats and Lyons quarry fugitive dust sources and raw materials dryer (P002) are not subject to the requirements in NSPS OOO because these activities are not affected facilities as identified in 60.670(e). In addition, the activities at P000 (raw material storage and handling at plant site) also are not affected facilities as identified in 60.670(e) and truck dumping of nonmetallic minerals into any screening operation, feed hopper, or crusher is exempt from the particulate matter requirements (see 60.672(c)). The primary (plant site) and secondary crushers are not subject to the requirements in NSPS OOO because they were constructed prior to August 31, 1983 and have not been modified.

The Dowe Flats primary crusher (AIRS pt 027) and belt conveyor (AIRS pt 026) are subject to the requirements in NSPS OOO and the appropriate requirements are included in the current permit (last revised November 17, 2017).

NSPS Subpart DDDD – Emission Guidelines for Commercial and Industrial Solid Waste Incineration (CISWI) Units that Commenced Construction on or Before November 30, 1999

The TRD for the second renewal permit (issues March 1, 2017) discusses the applicability of the CISWI requirements and they do not apply to the kiln. Please review the TRD for the second renewal permit for the detailed discussion. The permit includes language specifying that if CEMEX uses tires they must have been managed under the oversight of an established tire collection program or CISWI requirements would apply.

NSPS Subpart IIII – Stationary Compression Ignition Engines

NSPS Subpart IIII applies to stationary compression ignition engines that commenced construction, reconstruction or modification after July 11, 2005 and were manufactured after specified dates. The date the engine commenced construction is the date the engine was ordered by the owner/operator. The current permit (last revised November 17, 2017) includes four (4) diesel fuel-fired internal combustion engines. All but one of those engines commenced construction prior to July 11, 2005 and are not subject to the requirements in NSPS IIII.

On May 30, 2019, the source submitted an APEN for a 225 diesel fuel-fired engine (driving a 6" water pump) indicating that it was subject to the requirements of NSPS IIII. As discussed below, this engine was later determined to be a non-road engine. Non-

road engines are not stationary internal combustion engines and are therefore not subject to the requirements in NSPS IIII.

In addition, the Division's 2019 inspection report identifies a diesel fuel-fired engine, rated at 84 hp, driving a 8" water pump operating at the facility. The source submitted information indicating that this engine was manufactured prior to 1997. Therefore, the requirements in NSPS IIII would not apply to this engine.

NSPS Subpart JJJJ – Stationary Spark Ignition Engines

NSPS Subpart JJJJ applies to stationary spark ignition engines that commenced construction, reconstruction or modification after June 12, 2006 and were manufactured after specified dates. The date the engine commenced construction is the date the engine was ordered by the owner/operator. The current permit (last revised November 17, 2017) includes a natural gas-fired engine (used to rotate the kiln during power outages), however, this engine commenced construction before June 12, 2006. The Division's 2019 inspection report identifies a gasoline-fired engine, rated at 10.7 hp, driving a 4" trash pump operating at the facility. The source indicated that this engine was manufactured in March 2016 and was located at the facility in May 2019. Therefore, the requirements in NSPS Subpart JJJJ apply to this engine.

Non-Road Engines

Non-road engines are not subject to the requirements in NSPS Subparts IIII or JJJJ, as well as MACT ZZZZ. Non-road engines are portable or transportable and by definition do not remain in a single location for more than 12 consecutive months or shorter period of time if the engine is located at a seasonal source. The definition of "stationary source" in Colorado Regulation No. 3, Part A, Section I.B.52, notes the following regarding non-road engines "[t]hose emissions resulting directly from an internal combustion engine for transportation purposes or from a non-road engine as defined in Section I.B.34 . of this Part shall not be considered a stationary source."

In their renewal application, the source indicated that the Dowe Flats dewatering pumps were "portable and transportable" and were moved at least once per year as part of their intended function. The source provided additional information on the function of these engines in an email noting that the engines "are used for general dewatering at the site, filling water trucks/wagons for dust control, as well as dewatering pits and water filled reclamation areas. They are moved throughout the site as needed."

Based on this additional information the Division agreed that the four Dowe Flats dewatering pumps were non-road engines. These four engines are diesel fuel-fired, two were included in the current permit (last revised November 17, 2017). Only one of these engines, the 225 hp engine, would have been subject to NSPS IIII if it was a stationary internal combustion engines. Since non-road engines are not stationary sources, they are not subject to permitting requirements except as provided for in Regulation No. 3, Part A, Section I.B.36.c and d, which does not apply to these engines. For informational

purposes, the Dowe Flats dewatering pumps have been included in the insignificant activity list in Appendix A of the permit.

Colorado Regulations

Regulation No. 1 - Emission Control for Particulate Matter, Smoke, Carbon Monoxide and Sulfur Oxides (Adopted 06/21/07, effective 08/30/07)

The following requirements apply to emission sources at this facility: opacity requirements (Section II.A.1 and 4), particulate matter requirements for manufacturing processes and fugitive particulate emissions (Sections III.C and D) and sulfur dioxide requirements (Sections VI.A.3.f (facility) and VI.B.4.b.(i) (diesel fuel-fired engines)).

Update following public comment period (October 12 – November 11, 2023, public comment hearing January 5, 2024): As provided for in Regulation No. 1, Section II.A.6.c, fugitive emission sources are not subject to the opacity requirements in Sections II.A.1 and 2. Therefore, fugitive emission sources are not subject to the Regulation No. 1 opacity limits. It should be noted that the current permit (last revised November 17, 2017) indicates that some of the emission sources associated with P000 (raw material storage and handling at plant site, Section II.3 of the current permit) were subject to opacity requirements. After further review following the public comment period for the draft permit, the Division considers that the emission sources associated with P000 (Section II.2 of the draft renewal permit) are all fugitive emission sources and are not subject to the Regulation No. 1 opacity requirements. **End of update.**

The particulate matter requirements for manufacturing processes were streamlined with respect to some equipment in favor of other particulate matter limitations as discussed in previous TRDs. It should also be noted that although fuel is burned in the raw materials dryer and the kiln, the particulate matter limits for manufacturing processes apply rather than the particulate matter limits for fuel burning equipment in accordance with a May 7, 1998 internal memo.

The current permit (last revised November 17, 2017) includes the SO₂ limit from Regulation No. 1 for existing cement manufacture (Section IV.A.3.f), this limit is included in Section II.10 of the permit which includes requirements for the kiln. The kiln is the primary source of SO₂ emissions at the facility, although SO₂ emissions would also be emitted by the raw materials dryer. Regulation No. 1 defines an existing source as “sources constructed or modified” prior to August 11, 1977 (see Regulation No. 1, Section VI.A) and a new source as “a newly constructed or modified source of sulfur dioxide emissions that has not been issued an Emission Permit (in accord with Regulation No. 3 of this commission) prior to the August 11, 1977 effective date of this amended regulation” (see Regulation No. 1, Section VI.B.1).

As discussed previously, this facility commenced operation in the 1968-1969 time frame, however, the kiln was modified in 1979-1980 to shorten it and replace the roaster (which was previously used to volatilize kerogens) with a flash vessel (which acts similar

to a pre-calciner). Prior to the kiln conversion, the roaster and dryer were considered one system. As a result, construction permits were issued for both the dryer and kiln on October 18, 1979. Thus the kiln and dryer were modified after August 11, 1977, had not been issued an emission permit prior to that date and could have been considered subject to the requirements in Section VI.B.5 for new or modified sources. It should be noted that the dryer was modified in 1985 to burn coal and then modified in 1993 to revert to back to burning natural gas as the primary fuel.

Regulation No. 1, Section VI.B.3 states the following regarding modifications:

The term “modification” is as defined in the Common Provisions Regulation, Section I.G. except that any source of sulfur dioxide subject to an emission standard which measures the sum of all sulfur dioxide emissions from a given facility shall not be considered “modified” for the purposes of this regulation unless the alteration may cause an increase in the sum of all sulfur dioxide emissions from such facility.

The Common Provisions Regulation no longer includes the definition of modification, it is included in Regulation No. 3, Part A, Section I B. 28, but the above language from Reg 1 provides special provisions for sources subject to a limit on the sum of all sulfur dioxide emissions from a given facility. The modifications in 1979-1980 for the kiln and dryer were not anticipated to result in an increase in SO₂ emissions, thus would not be considered a modification in accordance with the provisions in Reg 1, Section VI.B.3. The modification to the dryer in 1985 (to convert to burning coal) could have resulted in an increase in SO₂ emissions, although it was thought that SO₂ emissions would decrease due to interaction with the raw material (primarily limestone). In 1993 the dryer was revised to revert to burning natural gas as the primary fuel. Since no modifications would have led to “an increase in the sum of all sulfur dioxide emissions from such facility” the Division considers that the facility is appropriately subject to the Reg 1 requirements for existing sources in Section IV.A.3.f, which is included in the current permit (last revised November 17, 2017).

As noted previously the SO₂ requirements in IV.A.3.f apply to SO₂ emissions from the facility, yet the limit is only included in the kiln section of the permit. Since the dryer and the kiln are the primary sources of SO₂ emissions from the kiln, the SO₂ requirement will also be included in Section II.5 of the permit as well. Note that other sources of SO₂ emissions include engines at this facility. However since the engines use low sulfur fuels, are small and not operated frequently, SO₂ emissions are anticipated to be very low. As a result the permit does not specify that SO₂ emissions from the engine be included in assessing compliance with the Reg 1 SO₂ limit.

Colorado Regulation No. 6, Part B – Standards of Performance for New Stationary Sources (Non-Federal) – **State-Only Requirements** (Adopted 5/20/21, effective 7/15/21)

Regulation No. 6, Part B includes non-federal (state-only) requirements for various categories of new sources. For this facility, manufacturing processes constructed, modified or reconstructed after January 30, 1979 are subject to requirements in Section III.

As discussed previously, this facility commenced operation in the 1968-1969 time frame, although some of the equipment has been modified. The following equipment was constructed, reconstructed or modified after January 30, 1979 and is potentially subject to the requirements in Regulation No. 6, Part B, Section III):

- P017 (Dowe Flats and Lyons quarry – fugitive dust sources)
- P017 (Dowe Flats quarry – point source emissions – primary crusher (AIRS pt 027) and belt conveyor (AIRS pt 026))
- P000 (raw material storage and handling at plant site)
- P002 (raw materials drying)
- P007 (kiln burning)
- P009 (clinker and gypsum/additive silos and weigh feeders)
- P010 (sheltered (A-Frame) clinker storage and reclamation)
- P015 (outdoor clinker piles and handling)
- P011 & P012 (cement finish mill and auxiliaries)
- P013 (cement silos/packhouse/loading)
- P007A (handling and processing of CKD and raw material waste dust)
- P050 (cement rail car unloading system)

The activities addressed under Dowe Flats and Lyons quarry fugitive particulate matter sources are not considered manufacturing processes and are not subject to the requirements in Regulation No. 6, Part B, Section III. The remaining sources are subject to both the particulate matter and opacity requirements in Section III.C and as discussed in previous TRDs, these requirements were streamlined in favor of other particulate matter and opacity requirements. The previous determinations for streamlining those requirements are still appropriate, with one exception.

The reason for streamlining the Reg 6, Part B 20% opacity requirement was because the equipment was subject to a more stringent opacity limit in either NSPS F or OOO (10% or 7%). This was correct for all equipment/activities except for P000 (raw materials storage and handling at plant site). In the initial Title V permit (issued February 1, 2000), the Division considered that this equipment associated with P000 (previously labeled as P004) was subject to requirements in NSPS F (see page 6 of TRD for initial permit (issued February 1, 2000)). When the first renewal permit was processed, the Division noted that the activities in P000 (raw materials storage and handling at plant site, included in Section II.3 of the permit) were not subject to either NSPS F or OOO (see page 2 of the TRD for the first renewal (issued March 1, 2008)). Thus streamlining the Reg 6, Part B opacity requirements was not appropriate and the permit will be revised to reflect that.

Update following public comment period (October 12 – November 11, 2023, public comment hearing January 5, 2024): After further review, the Division determined that the activities addressed under P000 (raw material storage and handling at plant site) are not considered manufacturing processes and are not subject to the requirements in Regulation No. 6, Part B, Section III. The Common Provisions Regulation (5-CCR-1001-2) defines “manufacturing process or process equipment” as follows:

An action, operation, or treatment involving chemical, industrial, or manufacturing factors, such as heat treating furnaces, or fuel-burning devices that are a part of a manufacturing process where emissions are intermixed with the process emissions, heating and reheating furnaces, sintering trains, electric furnaces, kilns, dryers, roasters, painting ovens, direct fired drying ovens, crushers, and all other methods and forms of manufacturing or processing that emit, or affect the emission of air pollutants, but not including fuel-burning equipment.

The activities addressed under P000 (truck unloading, front end loading activities and unloading at storage piles or crusher feed) include material storage and handling activities prior to entering a manufacturing process (e.g. the crusher). Thus they are not subject to the requirements in Section III. **End of update.**

In addition, it should be noted that up until the April 1, 2013 issuance of this permit (first renewal, modification 1), the permit shield for streamlined conditions (Section III.3) included the Reg 6, Part B 20% opacity limit for the raw material dryer (P002). This was an inadvertent error and is being corrected at this time. It should also be noted that previous versions of the permit did not include the Reg 6, Part B, PM limits for the raw material dryer in the permit shield for streamlined conditions (Section III.3). The Division also considers that this is likely an error which is also being corrected at this time.

As previously noted, the kiln (P007) and the raw materials dryer (P002) are also sources of SO₂ emissions and Reg 6, Part B, Section III.D includes SO₂ emission limitations. The kiln and the dryer are sources of SO₂ emissions and although both were noted as being subject to the Reg 6, Part B PM and/or opacity limits (Reg 6, Part B requirements are included in the table for streamlined conditions in Section III of the current permit (last revised November 17, 2017)) the Reg 6, Part B SO₂ limits were not addressed in the permit or discussed in previous TRDs. Reg 6, Part B, Section I, adopts the general provision in 40 CFR Part 60 Subpart A (Part B, Section I.A adopts the general provisions specified in Part A, which adopts 40 CFR Part 60 Subpart A by reference). Under the modification definition in 60.permit) and 60.14(a) notes the following (emphasis added):

Upon modification, an existing facility shall become an affected facility for each pollutant to which a standard applies and for which there is an increase in the emission rate to the atmosphere.

Thus a modification can trigger NSPS requirements for some but not all regulated pollutants. As discussed previously under Regulation No. 1, the modifications to the kiln and dryer did not result in an increase in SO₂ emissions, therefore the Reg, Part B Section III.D SO₂ requirements were not triggered.

Colorado Regulation No. 3 – Stationary Source Permitting and Air Pollutant Emission Notice Requirements

Revisions to Regulation No. 3 were adopted by the Air Quality Control Commission (AQCC) on December 15, 2022, effective February 14, 2023, to require additional APEN reporting requirements and fees for certain sources.

Specifically the revisions require the submittal of a facility-wide greenhouse gas (GHG) emissions APEN for sources required to report GHG emissions in accordance with Regulation No. 22. The GHG APEN is required to be submitted on or before December 31, 2023 (for 2022 calendar year emissions) and annually thereafter. This is a state-only requirement.

In addition major stationary sources of VOC and/or NO_x (potential to emit NO_x and/or VOC greater than or equal to 25 tons/yr) located in the ozone nonattainment area are required to submit annual statements to certify that annual, actual VOC and/or NO_x emissions are as reported on APENs. Initial certifications are to be submitted by January 31, 2024 and annually, thereafter.

Finally, the revisions include fees for major stationary sources of NO_x and/or VOC (potential to emit NO_x and/or VOC greater than or equal to 25 tons/yr) located in the 8-hr Ozone Control Area in the event that the 2008 ozone standard is not met by the deadline (i.e. 2027, unless an extension is granted).

These new requirements apply to this facility and as such will be included in the permit. Currently, APEN reporting requirements and fees are included in the General Permit Conditions (Section IV of the permit), therefore, as discussed later in this document, the General Permit Conditions will be revised to address these new requirements.

Colorado Regulation No. 7 - Control of Ozone via Ozone Precursors and Control of Hydrocarbons via Oil and Gas Emissions (Emissions of Volatile Organic Compounds and Nitrogen Oxides) (Adopted 12/15/2022, effective 2/14/2023)

The TRD for the second renewal permit (issued March 1, 2017) addressed the applicability requirements in Regulation No. 7. Since that time, there have been numerous revisions to Regulation No. 7, including a reorganization that added parts (e.g. Part A) and revised section numbers within those parts. The below discussion addresses the version of Regulation No. 7 adopted by the AQCC on December 15, 2022, effective February 14, 2023.

Note that at the April 19-21, 2023 meeting, the AQCC reorganized Regulation No. 7 into four separate regulations. The provisions in Part B become Regulation No. 24, Part C become Regulation No. 25, Part D would remain Regulation No. 7 and Part E become Regulation No. 26

The provisions in Part A “Applicability and General Provisions” include no actionable applicable requirements.

The provisions in Part B “Storage, Transfer, and Disposal of Volatile Organic Compounds and Petroleum Liquids and Petroleum Processing and Refining” in general do not apply to this facility. However Part B includes general requirements for the storage, transfer and disposal of volatile organic compounds (Sections I and III) which are included in the General Permit Conditions (Section IV). In addition, there is a gasoline storage tank which is subject to requirements in Section IV.B.3. As previously noted, these requirements were included in “new” Regulation No. 24 following the April 19-21, 2023 AQCC meeting.

The provisions in Part C “Surface Coating, Solvents, Asphalt, Graphic Arts and Printing, and Pharmaceuticals” in general do not apply to this facility. However, the permit includes requirements for solvents, specifically requirements for transfer and storage of waste/used solvent (Sections II.A.3 and 4), as well as requirements for solvent cold cleaners (Section II.B). Since the last renewal, additional solvent requirements potentially apply if actual, uncontrolled emissions exceed either two (2) or three (3) tons in any calendar year. The permit will include language indicating if such levels are exceeded certain requirements apply. As previously noted, these requirements were included in “new” Regulation No. 25 following the April 19-21, 2023 AQCC meeting.

The provisions in Part D “Oil and Natural Gas Operations” do not apply to this facility.

The provisions in Part E “Combustion Equipment and Major Source RACT” potentially apply to equipment at this facility. The engine requirements in Section I apply to natural gas-fired engines. The only natural gas-fired engine at this facility is less than 500 hp and has actual, uncontrolled emissions below the APEN de minimis level. Therefore, the engine is not subject to any engine requirements in Section I. The dryer is subject to the combustion process adjustment requirements in Section II.A.6 and those requirements will be included in the permit. Since the kiln does not meet the definition of a “lightweight aggregate kiln” and the diesel fuel-fired engines are all less than 500 hp, the emission limitations and compliance demonstration requirements in Section II.A.4 and 5 do not apply. The facility is subject to requirements in Section III.A.2, which requires the facility to comply with the applicable total hydrocarbon (THC) requirements in the 40 CFR Part 63, Subpart LLL. This requirements will be streamlined in favor of the requirements in 40 CFR Part 63 Subpart LLL which are included in the current permit. As previously noted, these requirements were included in “new” Regulation No. 26 following the April 19-21, 2023 AQCC meeting.

Colorado Regulation No. 22 – Colorado Greenhouse Gas Reporting and Emission Reduction Requirements

The GHG reporting requirements in Regulation No. 22, Part A apply to any facility that is required to report under the federal GHG reporting requirements in 40 CFR Part 98 (see Reg 22, Part A, Section III.A.6). Since this facility contains a source category listed in Table A-3 (cement manufacture), is it required to report GHG emissions under the federal GHG reporting rule and is subject to Colorado’s GHG reporting requirements. Neither the federal nor Colorado’s GHG reporting requirements meet the definition of an applicable requirement (see Reg 3, Part A, Section II.B.9), therefore, they are not required to be included in the Title V permit. Additionally, Regulation No. 22 does not specify that the GHG reporting requirements be included in any air permit. Therefore, the GHG reporting requirements in Reg 22, Part A will not be included in this renewal permit. Note that as discussed previously, Reg 3 was recently revised to require facility-wide reporting of GHG emission on an APEN. Those requirements apply to this facility and will be included in the permit.

Regulation No. 22, Part B includes requirements to reduce GHG emissions and this facility is subject to the requirements in Section II (Greenhouse Gas Emissions and Energy Management for the Manufacturing (GEMM) Sector in Colorado, otherwise referred to as “GEMM”. This facility is currently undergoing the process to establish GHG emission limitations at this time. The process begins with the audit report that was submitted by the December 31, 2022 deadline. The Division took comments on the audit report during January 2023. Audits are required every five years, with the final audit to be completed in 2037.

Within 60 days of receipt of the audit report, the Division is required to determine GHG best available emission control technology (BAECT) and energy best management practices (BMPs). The Division is required to hold one or more public meetings on the final GHG BAECT and energy BMP. Within 45 days of making its final GHG BAECT and energy BMP determinations the Division is required to present the determinations to the AQCC. Within 120 days of AQCC approval of the GHG BAECT and energy BMPs, the source is required to submit a compliance action plan to the Division for review and approval. Beginning no later than the third year after each audit year the source must demonstrate a mass-based five (5) percent reduction in GHG emissions. Beginning on May 1, 2026, the source is required to submit an annual report that includes an annual compliance certification.

Any GHG emission limitation that is established under the provisions in Reg 22, Part B, Section III does not meet the definition of an applicable requirement (see Reg 3, Part A, Section II.B.9), therefore, it would not have to be included in the Title V permit. Additionally, Reg 22, Part B, Section II does not specify that GHG emission limits established under those provisions be included in any air permit. Therefore, any GHG emission limits established for this facility under the provisions of Reg 22, Part B, Section II will not be included in this renewal permit.

Note that in the April 19-21, 2023 meeting, the AQCC moved the GHG reduction requirements from the manufacturing section (Regulation No. 22, Section II.B) into “new” regulation No. 27.

Colorado Regulation No. 23 – Regional Haze Limits

The RH requirements in Regulation No. 23 were previously included in Regulation No. 3, Part F. The AQCC adopted new emission reduction requirements on December 17, 2021 (effective January 30, 2022) to meet or exceed the reasonable progress goals during the second 10-year planning period. The second 10-year planning period addresses other sources of emissions at the facility than were previously addressed during the best available retrofit technology (BART) and RP determinations made in 2011 and included in Regulation No. 3, Part F.

During the second 10-year planning period, the following emission sources were evaluated: the kiln, raw materials grinding and the Dowe Flats Quarry. For the second planning period, only SO₂ and NO_x emissions from the kiln were reviewed. The SO₂ emission limitations, as well as the annual (tons/yr) NO_x emission limit remain the same as the 2011 determination, however, an additional short-term NO_x limit was included. As discussed later in this document, the permit will include the appropriate requirements from Regulation No. 23.

Statutory Requirements

The Colorado legislature has taken action in recent years to address emissions from air toxics. These requirements have been included in Colorado’s Revised Statutes (C.R.S.) but have not been codified in Colorado Regulations. The recent legislative action regarding air toxics emissions that apply to this facility is the requirement in HB22-1244 to report toxic emissions annually. Specifically C.R.S. 25-7-109.5(4)(a) requires synthetic minor sources and sources required to have an operating permit to submit annual air toxic reports to the Division. The reports are to be submitted beginning June 30, 2024 and shall cover the previous calendar year. The requirement to submit annual air toxic reports does not meet the definition of an applicable requirement (see Reg 3, Part A, Section II.B.9), therefore, it would not have to be included in the Title V permit. In addition, the language in the statute does not specify that the requirement to submit annual air toxic reports be included in any air permit. Therefore, the requirement to submit an annual air toxic report will not be included in this renewal permit.

Consent Decree Requirements

As discussed in the TRD for the second renewal (issued March 1, 2017), a Consent Decree (09-cv-0019-MSK-MEH) was lodged in the federal District Court on April 19, 2013 related to NO_x emissions at this facility. The Consent Decree (CD) specified that certain requirements be included in a permit and those requirements were included in the second renewal permit (issued March 1, 2017).

V. New Emission Sources

The source submitted an APEN on May 30, 2019 for a diesel fuel-fired internal combustion engine, rated at 225 hp, which is not addressed in the current permit (last revised November 17, 2017). This engine drives a 6" water pump used at Dowe Flats.

In addition, a review of the Division's 2019 inspection report indicates two additional engines operating at the facility that are not currently included in the permit. One of these engines (84 hp, diesel fuel-fired) drives an 8" water pump used at Dowe Flats, the other (10.7 hp, gasoline-fired) drives a 4" trash pump.

As discussed previously in Section IV (Applicable Requirements) on page 14, the source considers that the Dowe Flats water pump engines (225 hp and 84 hp diesel fuel-fired) meet the definition of non-road engines. The Division has agreed and will include these engines in the insignificant activity list in Appendix A of the permit.

The 4" trash pump engine does not meet the definition of a non-road engine. This engine is subject to requirements in 40 CFR Part 60 Subpart JJJJ and 40 CFR Part 63 Subpart ZZZZ. Therefore, regardless of the level of emissions, this engine must be included in the permit under the "catch-all" provisions in Colorado Regulation No. 3, Part C, Section II.E.

Trash Pump, EGEN-DF-5, Honda, Model No. GX340, rated at 10.7 hp, serial no. not available. This engine burns gasoline, maximum fuel use rated at 0.92 gal/hr. Manufactured in March 2016. It commenced operation at the facility in May 2019 and had not operated prior to operation at the facility.

1. Applicable Requirements

Based on the information provided by the source and available to the Division, the engine is subject to the following applicable requirements.

- Except as provided for below, visible emissions shall not exceed 20% opacity (Reg 1, Section II.A.1)
- Visible emissions shall not exceed 30% opacity, for a period or periods aggregating more than six (6) minutes in any sixty (60) minute period, during fire building, cleaning of fire boxes, soot blowing, start-up, process modifications, or adjustment or occasional cleaning of control equipment (Reg 1, Section II.A.4)

Based on engineering judgment, the Division believes that the operational activities of fire building, cleaning of fire boxes and soot blowing do not apply to gasoline-fired engines. In addition, since this engine is not equipped with control equipment the operational activities of adjustment or occasional cleaning of control equipment do not apply to it. Finally, based on engineering judgment, it is unlikely that process modifications will occur with this engine. Therefore, for this

unit the 30% opacity provision only applies during startup. The 20% opacity requirement (noted in the above bullet) applies at all other times. Note 40 CFR Part 63 Subpart ZZZZ § 63.6625(h) specifies that startup shall not exceed 30 minutes.

- Emission and throughput limits

Since actual emissions from this engine are below the APEN de minimis level, no emission limits will be included in the permit. The source will be required to record hours of operation and calculate emissions annually to determine if emissions exceed the APEN de minimis level.

- 40 CFR Part 60 Subpart JJJJ requirements

The 10.7 hp gasoline fired engine is subject to the NSPS JJJJ requirements, which includes emission limitations, as well as operating requirements.

The 10.7 hp engine is non-handheld and has an engine displacement greater than 225 cubic centimeters and is subject to the phase 3, Class II engine exhaust requirements in 40 CFR Part 1054 § 1054.105(a).

- 40 CFR Part 60 Subpart A, general provisions

- Circumvention (§ 60.12)

40 CFR Part 60 Subpart JJJJ § 60.4246 identifies the general provisions that apply. According to the table, the provisions in § 60.7 (notification and recordkeeping) apply as specified in § 60.4245 and this section does not apply to this engine, therefore, the provisions in § 60.7 do not apply. The table also indicates that § 60.8 (performance testing) only applies to engines subject to performance testing requirements and therefore does not apply to this engine. In addition, the table indicates that the provisions in § 60.11 do not apply as the requirements are specified in Subpart JJJJ. In addition §§ 60.13 (monitoring requirements) and 60.18 (general control requirements) do not apply.

- 40 CFR Part 63 Subpart ZZZZ requirements

This engine meets the Subpart ZZZZ requirements by meeting the NSPS JJJJ requirements.

Since this engine meets the requirements in Subpart ZZZZ by meeting the requirements in NSPS JJJJ, the general provisions in 40 CFR Part 63 Subpart A also do not apply.

2. Emission Factors

Potential to emit from the new engine is shown in the table below. Emissions were estimated using the listed emission factors, maximum power (kW) and year-round operation (8,760 hours).

Pollutant	Emission Factor ¹ (lb/hp-hr)	Emissions ² (tons/yr)	Significance Level ³ (ton/yr)
PM	7.20E-04	0.03	25
PM ₁₀	7.20E-04	0.03	15
PM _{2.5}	7.20E-04	0.03	10
SO ₂	7.20E-04	0.03	40
NO _x	0.011	0.52 [0.62]	40
CO	6.90E-03	0.32 [47.00]	100
VOC ⁴	0.02	0.94	40

¹Emission factors from AP-42 - Section 3.3 (dated 10/1996), Table 3.3-1, for gasoline engines < 250 hp and diesel engines < 600 hp

²Emissions in “[]” based on NSPS IIII or JJJJ limits.

³Indicates the NANSR significance level on the date the engine was installed (2019). The area was classified as a serious ozone non-attainment area on January 27, 2020. The significance level dropped to 25 tons/yr of VOC or NO_x on January 27, 2020. The area has since been classified as a severe ozone non-attainment area on November 7, 2022.

⁴Emission factor for total organic compounds (TOC) exhaust plus crankcase.

The trash pump engine has actual emissions below the APEN de minimis level and potential emissions below the significance level at the time it was located at the facility. Potential emissions are also below the current significance level (25 tons/yr of VOC or NO_x). Therefore, emission limits have not been included in the permit for this engine.

3. Monitoring Plan

The source will be required to comply with the NSPS JJJJ requirements.

In addition, the permit will require that hours of operation be recorded and emissions calculated to determine if an APEN must be filed.

Finally, the permit will require that a Method 9 opacity observation be conducted annually (calendar year) if the engine is operated during the calendar year.

4. Compliance Status

In the absence of credible evidence to the contrary, this engine is presumed to be in compliance with all applicable requirements.

VI. Modifications Requested by the Source

The following discussion related to modifications is with respect to the current active permit (last revised November 17, 2017) and unless specifically noted as “new”, the condition numbers identified in this document reflect the condition numbers in the current permit (last revised November 17, 2017). Because some permit conditions in the current Title V permit have been removed, reorganized and/or reformatted as part of this permitting process, the condition numbers discussed in this document may not reflect the condition numbers in the draft Title V permit.

May 30, 2019 Submittal

The source submitted an APEN on May 30, 2019 for an engine (225 hp) that was not previously reported at the facility. Based on the source’s comments on the draft permit received on August 31, 2023 and a follow-up email, the Division agrees that this engine meets the definition of a non-road engine and is not required to be addressed in the permit because it is not a stationary source (see previous discussion in Section IV (Applicable Requirements) on page 14). For information purposes, the engine has been included in the insignificant activity list in Appendix A of the permit.

March 1, 2021 Renewal Application

The renewal application included the appropriate application forms, a cover letter describing some specific changes and a red-lined permit noting additional changes. The following changes noted in the renewal application were made as discussed below:

Changes Made:

Page Following Cover Page

- Changed the mailing address under “issued to.”

General

- Revised the citations for the RH requirements from Regulation No. 3, Part F to Regulation No. 23.

Section I – General Activities and Summary

- Revised the description in Condition 1.1 as requested.
- The following changes were made to the table in Condition 4.1:
 - Replaced references to “PM Emission Control Plan” with “Fugitive Particulate Emission Control Plan.”
 - Under the column “Pollution Control Device” corrected the number of baghouses for P013 (AIRS pt 013) to 6.

- Moved the language under “Plant ID” to equipment description for the stationary internal combustion engines and included the engine id numbers under “Plant ID)
- Removed the Dowe Flats 6” and 8” pump engines.

Section II.1 – P017 – Dowe Flats and Lyons Quarry – Fugitive Dust Source

- Revised the section header to replace “dust” with “particulate emissions”.
- Condition 1.6 in the summary table was revised to say “Fugitive Particulate Emission Control Plan.”
- Made some corrections to the note under Conditions 1.3.
- Replaced “insure” with “ensure” in Condition 1.6.1.4.

Section II.2 – Dowe Flats Quarry – Point Source Emissions

- Added language to Condition 2.5 to indicate that if the equipment is not operating on the required date of the performance test, the test must be conducted within 60 calendar days of startup.

Section II.3 –Raw Material Storage and Handling at Plant Site

- Condition 3.5 in the summary table was revised to say “Fugitive Particulate Emission Control Plan.”

Section II.5 – Raw Materials Drying

- Revised Conditions 5.6.2 and 5.13.3.1 to remove the beginning phrases related to the compliance date. Note that Condition 5.13.3.1 is renumbered as Condition 5.7.1. As discussed later in this document, the RH NO_x and SO₂ limits and monitoring requirements in Condition 5.13 were moved to Condition 5.7.
- Removed Condition 15.3.2 (RH compliance date). Note that the remainder of Condition 15.3 was removed as discussed later in this document.

Section II.8 – Raw Material Grinding

- Corrected the typo in Condition 8.3.

Section II.10 – Kiln Burning

- Removed Condition 10.22.2.2 (RH compliance date). Note that the remainder of Condition 10.22 was removed as discussed later in this document.

Section II.14 – General Fugitive Particulate Emissions Requirements

- Revised the section header to include “particulate” after “emissions”.

Section II.16 - Opacity – Daily Plant Walk Through

- Capitalized “ckd” in Condition 16.1 and replaced “insure” with “ensure” in Condition 16.1.2.1

Section II.22 – 40 CFR Part 63 Subpart LLL Requirements

- Corrected the dioxin-furan (DF) limit in the Table in Condition 22.4.

Section II.26 – Stationary Internal Combustion Engines

- Removed the Dowe Flats 6” and 8” pump from this section of the permit.

Appendices

- Included the Dowe Flats 6” and 8” pumps in the insignificant activity list in Appendix A.
- Removed the Dowe Flats 6” and 8” pumps from the tables in Appendices B and C.

Changes Not Made:

The following changes requested in the renewal application were not made for the reasons noted.

Change “PM Emission Control Plan” references to “Fugitive Dust Control Plan”

In the cover letter for the renewal application, the source requested a global change to replace “PM Emission Control Plan” with “Fugitive Dust Control Plan” and the red-lined permit noted several of these corrections. The requirement for the PM emission control plan is in Colorado Regulation No. 1, Section III.D and certain provisions in that section require “fugitive particulate emission control plans,” thus making CEMEX’s suggested changes would be inconsistent with the language in Regulation No. 1. Therefore, the suggested changes will not be made. Note however, the Division revised the language in the table in Section I, Condition 4.1 to replace “PM Emission Control Plan” with “Fugitive Particulate Emission Control Plan” to be consistent with the language in Regulation No. 1.

Remove the Continuous Opacity Monitoring Systems (COMS) on the Kiln and Clinker Cooler

In the cover letter for the renewal application, the source requested that the requirement to operate COMS for both the kiln and clinker cooler be removed. Additional information supporting this request was included in Attachment 2 of the renewal application. The information supporting the request contends that using the particulate matter (PM)

continuous parameter monitoring systems (CPMS) for the kiln and clinker cooler is sufficient to monitor compliance with the Colorado Regulation No. 1 opacity requirements. The PM CPMS are required by 40 CFR Part 63 Subpart LLL, National Emission Standards for Hazardous Air Pollutants from the Portland Cement Manufacturing Industry (otherwise known as the “PC MACT”).

Attachment 2 discusses background on the removal of the opacity requirements for the kiln and clinker cooler in the PC MACT. EPA considered opacity to be a surrogate for PM, thus when the PC MACT was revised to require the use of a PM CPMS or continuous emission monitoring system (CEMS) for assessing compliance with the PC MACT PM limits, EPA removed the opacity limits for the kiln and clinker cooler. EPA considered that the PM CPMS is considered to be more sensitive than a COMS for measuring particulate matter. The Division does not disagree that a PM CPMS is a more sensitive monitoring system for measuring PM emissions. However, the current permit for CEMEX requires that the kiln and clinker cooler COMS be used to monitor compliance with opacity limits in Colorado Regulation No. 1, a task for which the COMS are appropriately suited.

The source considers that compliance with the PM CPMS site specific operating limit in the PC MACT can simultaneously demonstrate compliance with the opacity limits in Colorado Regulation No. 1. The site-specific PM CPMS operating limit is based on a 30-day rolling average, while the Colorado Regulation No. 1 opacity requirements are based on a 6-minute average. Thus using the PM monitoring specified in the PC MACT to assure compliance with the Regulation No. 1 opacity limits is not appropriate.

In addition, as indicated on page 48 of the TRD for the second renewal (issued March 1, 2017), the Division noted that the use of COMS for the kiln is required by regulation:

Under the RH requirements, the RH 20% opacity requirement does not apply if a PM CEMS is used to monitor compliance with the PM limit (see Regulation No. 3, Part F, Section VII.C.2.a). The PC MACT was revised on February 12, 2013 to require performance testing in conjunction with a PM CPMS to monitor compliance with the PM limit. Under the February 12, 2013 revisions kilns were still not subject to an opacity limit.

Under the RH requirements, if a PM CEMS is not required, then the opacity limit applies and a continuous opacity monitoring system (COMS) shall be used. Since the PC MACT no longer requires a PM CEMS but relies on performance tests and a PM CPMS, the RH opacity requirement does apply and a COMS shall be used.

The Division will continue to require that COMS be used for the kiln and clinker cooler in order to assess compliance with the opacity limits in the permit.

Remove Particulate Matter Performance Test Requirements

In the cover letter for the renewal application, the source requested that PM performance test requirements in Sections II.2, II.11 and II.13 of the permit be removed. The source indicates that the requirement to conduct stack tests for are not specified in any underlying regulations or requirements. Colorado Regulation No. 3, Part C, Section V.C.5.b states the following with respect to monitoring that must be included in a Title V permit:

Where the applicable requirement does not require periodic testing or instrumental or non-instrumental monitoring, periodic monitoring, or record keeping, sufficient to yield reliable data for the relevant time period that are representative of the source's compliance with the permit as required to be reported pursuant to Section V.C.16.e. of this Part C. Such monitoring requirements shall assure use of terms, test methods, units, averaging periods, and other statistical conventions consistent with the applicable requirements.

For these sources, the performance test requirements, in addition to the daily visible emission observations and baghouse operation and maintenance requirements were determined to be sufficient periodic monitoring requirements.

The sources in Sections II.11 and II.13 are subject to annual PM emission limits. The underlying construction permits for these sources set emission limits that kept PM₁₀ emissions from the project from exceeding the significance level and avoided NANSR requirements (the area was designated as a PM₁₀ non-attainment area at that time). Thus the Division considers that ongoing testing is appropriate for these sources.

The equipment in Section II.2 (Dowe Flats Quarry Crusher and Conveyor) is subject to NSPS OOO requirements and the Division has typically required performance tests every five years as periodic monitoring for sources subject to NSPS emission limitations, when the NSPS only required a one-time compliance test. Thus the Division considers that ongoing testing is appropriate for these sources. The Division acknowledges that these source may be operated less frequently, since raw materials are no longer being extracted from the Dowe Flats Quarry, thus the Division will add language in the permit to address this.

Remove Lead Performance Test Requirements for Dryer and Kiln

In the cover letter for the renewal application, the source requested that the lead performance test requirement for the Raw Materials Dryer be removed from the permit. The red-lined permit submitted with the renewal application suggested removal of the lead emission limit for the dryer, as well as suggesting removal of the lead testing requirement for the kiln. The Division will not remove the lead limit for the dryer, nor will the lead performance test requirements be removed for the dryer or kiln. For the dryer, the permit requires that performance tests be conducted every five years for lead, as well as SO₂, NO_x and CO. Although lead emissions from recent performance tests are

below the annual emission limitation, the Division does not consider it overly burdensome to test for lead in conjunction with other pollutants. Therefore, the lead performance test requirement for the dryer will remain in the permit.

For the kiln, the permit requires performance testing annually for PM and every 30 months for HCl, VOC and total hydrocarbons (THC). Although lead emission from recent performance tests for the kiln indicate emissions are well below the annual limitation, the Division does not consider it overly burdensome to test for lead in conjunction with other pollutants. Therefore, the lead performance test requirement for the kiln will remain in the permit.

November 15, 2022 Modification

The purpose of the November 15, 2022 modification was to remove the limestone throughput limit for P011 (AIRS pt 011). The requested change was addressed as follows:

Section II.11 – P009, P010, P015, P011, P012 and P013 – Various Material Handling Sources

- Removed the limestone limit for P011 in the summary table.

May 9, 2023 Submittal

The source submitted Title V application forms to update contacts listed in the permit. The current permit (last revised November 17, 2017) lists the Responsible Official and Facility Contact Person in the permit (on the page following the cover page). As noted in Section VII of this document, the responsible official and permit contact are no longer listed in the permit. Therefore, no changes to the permit will be made.

August 31, 2023 Comments on the Draft Permit and Technical Review Document

Based on the source's August 31, 2023 comments on the draft permit and technical review document, the following changes were made to the permit:

Note: The references to revised permit conditions in this particular section is with respect to the draft renewal permit, not the current permit (last revised November 17, 2017) as it is in the other sections of the document.

Section I – General Activities and Summary

- Minor revisions were made to the facility description in Condition 1.1 to reflect that the Dowe Flats crusher and conveyor are no longer in use.
- Removed construction permit number 94BO593 from the list in Condition 1.3.

- The following changes were made to the Table in Condition 4.1:
 - Removed Blasting (combustion byproduct emissions, AIRS pt 017 and “blasting” from description of AIRS pt 025) from the Dowe Flats Fugitive Sources.
 - Removed the Dowe Flats Point Sources (crusher and conveyor, AIRS pts 026 and 027).
 - Removed engines EGEN-DF-2 and EGEN-DF-4.

These engines were not in the current permit (last revised November 17, 2017) and were included in the draft renewal permit. However, as discussed previously, the Division has agreed that these are non-road engines and are not subject to permitting requirements.
 - Added a footnote to clarify that an APEN was filed for EGEN-LYO-1 (AIRS pt 053) but emissions were later determined to be below the APEN de minimis level and the APEN was cancelled.

Section II.1 – Dowe Flats and Lyons Quarry – Fugitive Particulate Emission Sources

- Removed AIRS pt 017 under the Section II.1 header.
- Removed the emission factors for blasting included in the table in Condition 1.2.
- Removed the ANFO limits in Condition 1.2. Conditions that follow are renumbered.
- Included as “new” Condition 1.4 a requirement indicating that blasting at Dowe Flats Quarry is prohibited.
- Removed the NO_x and CO emission limits in Condition 1.5. Conditions that follow are renumbered.
- The fugitive particulate emission control plan is included in Condition 1.6. (Note that with the removal of Conditions 1.2 and 1.4, it is renumbered as Condition 1.5.) Conditions 1.6.1.9, 1.6.1.12, 1.6.1.13, and 1.6.2.5 were removed since blasting at Dowe Flats is prohibited.
- Condition 1.10.1 (record number of blasts and material blasted) was removed, since blasting at Dowe Flats is prohibited.

Section II.2 – Dowe Flats Quarry Point Source Emissions

- This section of the permit was removed, since the crusher and conveyor are no longer operational and the underlying construction permit (94BO593) and AIRS pts (026 and 027) were cancelled. Conditions that follow are renumbered.

Section II.3 – Raw Materials Handling and Storage at Plant Site

Note that with the removal of Section II.2, this section is renumbered as Section II.2. The below discussion is based on the draft permit sent to the source for pre-public comment review.

- Revised Condition 3.1 to specify that throughput be determined daily based on truck deliveries and weight. Because there is a daily throughput limit and throughput is monitored based on trucks and truck weight, rather than belt scales, the Division considers daily monitoring is more appropriate.
- Revised Condition 3.2 to include specific recordkeeping requirements for number of truck deliveries and truck weights (empty and full).

Section II.16 – Daily Visible Emission Observations

Note that with the removal of Section II.2, this section is renumbered as Section II.15. The below discussion is based on the draft permit sent to the source for pre-public comment review.

- Removed “and conveyor” from the parenthetical language in Condition 16.1.
- Removed “NSPS OOO (Condition 2.2)” from the parenthetical language in Condition 16.5.

Section II.26 – Internal Combustion Engines

Note that with the removal of Section II.2, this section is renumbered as Section II.25. The below discussion is based on the draft permit sent to the source for pre-public comment review.

- Removed engines EGEN-DF-2 and EGEN-DF-4.

These engines were not in the current permit (last revised November 17, 2017) and were included in the draft renewal permit. However, as discussed previously, the Division has agreed that these are non-road engines and are not subject to permitting requirements.

- Revised Condition 26.3 to require that hours of operation be monitored and recorded annually. There are no longer sub-conditions under this requirement (e.g. 26.3.1)
- Removed the emission limit for EGEN-LYO-1 from Condition 26.4.2 and revised Condition 26.4 to require that emissions from all engines be calculated annually. With the revisions, Condition 26.4 no longer has sub-numbers (e.g. 26.4.1).

Section III – Permit Shield

- Made the following changes to the table in Section III.3 (streamlined conditions)

- Removed “Condition 2.2” from the column labeled “permit condition(s)”.
- Removed the reference to engine EGEN-DF-4 for Colorado Regulation No. 1, IV.B.4.b.(i) (SO₂) requirements from the column labeled “streamlined (subsumed) requirement”.

This engine was not in the current permit (last revised November 17, 2017) and were included in the draft renewal permit. However, as discussed previously, the Division has agreed that this is a non-road engine and is not subject to permitting requirements.

Appendices

- Included engine EGEN-DF-2 and EGEN-DF-4 in the insignificant activity list in Appendix A.

As previously noted, although these engines are not subject to permitting requirements, they will be included in the insignificant activity list for informational purposes.

- Made the following changes to the tables in Appendices B and C.
 - Removed the Dowe Flats Quarry – Point Source Emissions from the tables in Appendices B and C.
 - Removed engines EGEN-DF-2 and EGEN-DF-4.

These engines were not in the current permit (last revised November 17, 2017) and were included in the draft renewal permit. However, as discussed previously, the Division has agreed that these are non-road engines and are not subject to permitting requirements

- Removed the blasting emission factors from Appendix H since blasting is prohibited at Dowe Flats Quarry.

VII. Other Modifications

The following discussion related to modifications is with respect to the current active permit (last revised November 17, 2017) and unless specifically noted as “new”, the condition numbers identified in this document reflect the condition numbers in the current permit (last revised November 17, 2017). Because some permit conditions in the current Title V permit have been removed, reorganized and/or reformatted as part of this permitting process, the condition numbers discussed in this document may not reflect the condition numbers in the draft Title V permit.

In addition to the source requested modifications, the Division has included changes to make the permit more consistent with recently issued permits, include comments made by EPA on other Operating Permits, as well as correct errors or omissions identified during inspections and/or discrepancies identified during review of this renewal.

The Division has made the following revisions, based on recent internal permit processing decisions and EPA comments to the Lyons Cement Plant Renewal Operating Permit. These changes are as follows:

Page Following Cover Page

- Monitoring and compliance periods and report and certification due dates are shown as examples. The appropriate monitoring and compliance periods and report and certification due dates will be filled in after permit issuance and will be based on permit issuance date. Note that the source may request to keep the same monitoring and compliance periods and report and certification due dates as were provided in the previous permit. However, it should be noted that with this option, depending on the permit issuance date, the first monitoring period and compliance period may be short (i.e. less than 6 months and less than 1 year).
- Removed “Information Relied Upon” (renewal application and additional information received blocks), the Responsible Official and Facility Contact Person and the “note” under the submittal deadlines for the following reasons: the renewal application and additional information submittal dates are included in the TRD; listing individuals in the permit is not appropriate or necessary; and reports and compliance certifications are submitted electronically.

Section I – General Activities and Summary

- The citation for the definition of 8-hour ozone control area in Condition 1.1 was revised and a sentence was added indicating the 8-hr ozone control area has been classified as a severe ozone non-attainment area.
- Added “new” Section II, Condition 3.5 (20% opacity) and 10.1.1 (clinker monitoring) and Section IV, Conditions 22.f (GHG APEN reporting) and 30 (volatile organic compounds) as state-only requirements in Condition 1.4. Portions of the provisions in Section V, Condition 30 are state-only enforceable, so “(as noted)” was added after Condition 30. In addition, added a note indicating that the RH requirements are state-only until approved by EPA and that the provisions in Section IV, Conditions 3.d and 3.g are state-only on and after June 1, 2024.
- Revised the major stationary source threshold for NANSR in Condition 2.1 to 25 tons/yr. The threshold was lowered when the area was re-classified as a severe non-attainment area.
- The following changes were made to the table in Condition 4.1:
 - The column heading “AIRS ID” was revised to “AIRS Pt No.”.
 - Under the column “Process (Permit Section)” corrected the section numbers due to removal of some sections during the second renewal (issued March 1, 2017).

- Revised the description for P000 to remove “primary crushed”, since materials may not be crushed prior to storage on the patio.
- Revised descriptions of the engines to include the source’s identification numbers and to correct other information. Included AIRS point numbers as appropriate.
- Added the 4” trash pump (10.7 hp) engine to the table in Condition 4.1. The Division’s 2019 inspection report noted that these engines were not included in the permit.

Section II – General Changes

- References to emission factors from AP-42, Section 11.6 were updated to show the date AP-42, Section 11.6 was revised and to include the new table number. Note that although AP-42, Section 11.6 was revised, the emission factors included in the permit did not change with the latest AP-42 revision.

Section II.1 – Dowe Flats and Lyons Quarry – Fugitive Particulate Emission Sources

- The Dowe Flats Quarry is subject to a RH PM₁₀ emission limit, therefore, Condition 1.3 was revised to cite Regulation No. 23 as well as the construction permit.

Note that even though the RH PM₁₀ emission limit is the same as the construction permit limit, the RH limit was not streamlined but included in the citation because the RH emission limit cannot be revised without revising the state implementation plan (SIP). The construction permit limit was not streamlined since the RH limit is state-only enforceable until approved by EPA into Colorado’s Round 2 Regional Haze SIP. Thus both the CP and RH are cited. It should be noted that Regulation No. 23 does not include any recordkeeping, monitoring or reporting requirements for the Dowe Flats PM₁₀ emission limit.

Although the Lyons Quarry is listed in Regulation No. 23, Part A, Section IV.F.3, an emission limit is not included. The table in Section IV.F.3 simply states that “current permitted limit for Lyons Quarry below 10 TPY screening threshold”. Therefore, no changes to the permit are necessary to address the Lyons Quarry.

Section II.2 – Dowe Flats Quarry – Point Source Emissions

As discussed previously, this section was removed. Conditions that follow are renumbered.

Section II.3 – Raw Material Storage and Handling at Plant Site

Note that with the removal of Section II.2, this section is renumbered as Section II.2. The below discussion is based on the numbering in the current permit (last revised November 17, 2017).

- Revised the section header to remove “primary crushed” from the description for material discharged onto open stockpiles. Prior to closure of the Dowe Flats Quarry, material may have been crushed prior to being unloaded onto the patio.
- Revised the language in Condition 3.1 to indicate that materials discharged to storage on the plant patio and handled by front end loaders are subject to the throughput limits. The permit language currently implies the throughput limit only applies to the front end loaders.
- Included the Reg 6, Part B, Section III opacity requirement as “new” Condition 3.5. Note that in the current permit (last revised November 17, 2017), Condition 3.5 includes the fugitive particulate emission control plan requirements. Conditions that follow are re-numbered.

Section II.5 – Raw Materials Drying

Note that with the removal of Section II.2, this section is renumbered as Section II.4. The below discussion is based on the numbering in the current permit (last revised November 17, 2017).

- Revised Condition 5.1 to require that hours of operation be recorded daily. Daily hours of operation are required in order to determine daily SO₂ emissions and material processed for the Reg 1 facility wide SO₂ limit.
- Included a note in Condition 5.6.2 to indicate when the most recent PM performance test was conducted.
- Included the PM and PM₁₀ emission factors from the most recent test in the table in Condition 5.6.3.
- Condition 5.7 was split into two conditions 5.7 (SO₂ and NO_x) and 5.8 (CO, VOC and lead) in order to reflect the different underlying source of the requirements (SO₂ and NO_x from Reg. 23 and CO, VOC and lead from the construction permit). Note that as discussed below, the SO₂ and NO_x emission limits from the underlying construction permit will be streamlined (included in Section III.3 of the permit). Changes discussed below for Conditions 5.7.1 and 5.7.2 will be reflected in “new” Condition 5.8.
- Updated the paragraph in Condition 5.7.1 that starts with the phrase “a stack testing protocol shall” to reflect current language regarding test protocol, scheduling and report submittal.
- Updated Condition 5.7.1 to indicate the date of the last performance test.
- Updated the emission factors in the table in Condition 5.7.2 to include the results from the last performance test. The VOC and lead emission factors to be used for

periods when coal is used as fuel but a performance test when burning coal is not required were also updated to include the results from the most recent performance test. These factors were not updated with the last renewal permit (issued March 1, 2017).

The emission factors included for the 2021 test are based on the average of the 1st and 2nd runs, since the dryer feed and natural gas usage are significantly lower for the 3rd run. In addition, the dryer feed and natural gas usage rates during the 2021 test were lower than during the 2016 test, which further justifies using relying on the 1st and 2nd runs from the 2021 test.

- Removed Condition 5.7.3. As discussed below, the RH requirements for NO_x and SO₂ have been incorporated into Condition 5.7 and the NO_x and SO₂ emission limits from the construction permit have been streamlined (included in Section III.3 of the permit).
- Revised Condition 5.9 (VOC RACT) to update the citation.

Condition 5.9 includes the general RACT provisions in Regulation No. 7, for new or modified sources (this condition simply cites Section II.C – which addresses both new and existing sources). It should be noted that the requirement to conduct RACT for emissions from new or modified minor sources is also included in Regulation No. 3, Part B, Section III.D. The VOC RACT requirement was not included in the Construction Permit (12BO444-1) but was noted in the TRD (see page 11) for the initial permit (issued February 1, 2000) as a requirement that applied and as a result was included in the initial Title V permit. As previously discussed, Regulation No. 7 was revised in December 2019 and reorganized into parts (e.g. Part A) and then again in April 2023 to split it into separate regulations. Some of the general provisions in Regulation No. 7 (Part A) were retained in Regulation 7 and also included in the additional “new” regulations (24, 25 and 26) adopted April 20, 2023. Regulations No. 24 and 25 include the provisions in current Regulation No. 7, Part A, Section II.C for “new” and “existing” sources, while “new” Regulation No. 7 and 26 only include the provisions in Section II.C for “new” sources. Since the citation in Condition 5.9 this condition only identifies “Section II.C” of Regulation No. 7 (not specifying whether this applied to a “new” or “existing” source) and RACT is determined to be the design of the dryer (based a modification made in 1979-1980), the citation will be revised to indicate Regulation No. 24, Part A, Section II.C, which addresses both new and existing sources.

- The RH requirements in Condition 5.13 were removed. Since the SO₂ and NO_x RH emission limits are the same as the NO_x and SO₂ emission limits in the construction permit (included in Condition 5.7), the RH limits (Condition 5.13.1) and monitoring requirements (Condition 5.13.3) have been included in Condition 5.7 and the NO_x and SO₂ limits from the construction permit have been streamlined. The requirements in Condition 5.13.2 (compliance date) have passed and is no longer necessary.

- Included the Regulation No. 1 SO₂ requirements in Section VI.A.3.f as “new” Condition 5.14. This is a facility wide requirement that is currently only included in the permit for the kiln. For clarity, the facility wide requirement is being included for the dryer and language was added to specify the methodology to assess compliance with the facility wide emission limit.

This facility wide limit is 7 lb SO₂ per ton of material processed (including fuel). For purposes of assessing compliance with the limit, tons of material processed will be based on tons of feed to the kiln plus coal burned in the kiln and dryer (if applicable). Since any material processed through the dry would also be processed through the kiln, only feed material to the kiln will be included in the daily tons of material processed.

- The requirements in Colorado Regulation No. 7, Part E, Section II.A.6 were included in “new” Condition 5.15.

Note that the requirements in Section II.A.6.c.(ii) and the recordkeeping requirements in Section II.A.7.f.(iii) were not included because the dryer is not subject to any combustion process adjustment requirements in an NSPS or MACT. In addition, the recordkeeping requirements in Section II.A.7.f.(i)(F) were not include since the dryer is restricted to burning natural gas, except during periods of natural gas curtailment and emergencies.

As discussed previously, in April 2023 Regulation No. 7 was split into separate regulations. The provisions in Regulation No. 7, Part E were included in “new” Regulation No. 26. Therefore, the citations were revised to note that these requirements are now in Regulation No. 26.

Section II.8 – Raw Materials Grinding

Note that with the removal of Section II.2, this section is renumbered as Section II.7. The below discussion is based on the numbering in the current permit (last revised November 17, 2017).

- The emission factors identified in this section of the permit are included in Regulation No. 23 (Regional Haze Limits), Part A, Section IV.F.2 (this section states the reporting is based on the listed emission factors). Therefore, the permit has been revised to indicate that Regulation No. 23 requires the use of the PM emission factors for reporting purposes.

Section II.10 – Kiln

Note that with the removal of Section II.2, this section is renumbered as Section II.9. The below discussion is based on the numbering in the current permit (last revised November 17, 2017).

- Included the second sentence in Colorado Regulation No. 23, Part A, Section V.A.1.b.(ii) as “new” Condition 10.1.1.

The second sentence in Reg 23, Part A, Section V.A.1.b.(ii) requires that hourly clinker rates be monitored as required by 40 CFR Part 60 Subpart F § 60.63(b), which does not directly apply to the kiln. The source is required to monitor clinker production in accordance with the PC MACT requirements, specifically 63.1350(d), which is very similar to the NSPS F requirements in 60.63(b). However, 60.63(b) includes requirements to replace data for kiln operating hours for which there is no data on clinker production or the amount of feed to the kiln (see 60.63(b)(1)(iii)) and 63.1350(d) does not. As a result this portion of Reg 23, Part A, Section V.A.1.b.(ii) could not be streamlined.

- Included a “new” Condition 10.1.2 to require that daily clinker production rates be summed to get monthly clinker production and that monthly clinker production rates be summed to get the annual clinker production rate. Monthly and annual clinker production rates are used in calculating emissions.
- Included “new” Condition 10.1.2 to require that daily clinker production rates be summed to get monthly and annual clinker production rates to be used to calculate emissions.
- For conditions that specify that emission calculations use monthly or annual feed or clinker, added a reference to the permit condition that requires monitoring of the monthly or annual feed or clinker.
- Condition 10.5 was revised to include the CAM requirements as “new” Condition 10.5.4. In addition, Condition 10.5.1 was revised to specify that during each performance test a new CAM indicator range will be set (PM CPMS site-specific operating limit (SSOL)) and to require that a minor permit modification application be submitted to change the indicator range.
- The RH 20% opacity limit and monitoring language (previously included in Conditions 10.22.1.3 and 10.22.4.1) were included as “new” Conditions 10.9.3 and 10.9.8, conditions that follow were renumbered. In addition revised “new” Condition 10.9.4 (“old Condition 10.9.3) to replace “10.22.4.1” with “10.9.8”.

It is not clear whether the RH 20% opacity limit applies at all times. As noted on page 49 in the TRD for the second renewal (issued March 1, 2017) “Absent any information in the RH requirements that provides exceptions from the opacity requirement, the Division assumes that this RH opacity limit applies at all times”. Therefore the RH opacity requirement has been included in the permit.

- The following changes were made to the NO_x requirements in Condition 10.10 (summary table and text).

- Revised Condition 10.10.1 to include the annual (tons/yr) NO_x RH limit. The annual (tons/yr) NO_x limit from Construction Permit 12BO444-2 was streamlined (included in Section III.3 of the permit) since the RH limit is more stringent.
- The citation in Condition 10.10.2 was revised to replace the words “as provided for” with “Construction Permit 12BO444-2, as modified under”. The underlying construction permit, as well as the Title V permit were revised to incorporate the NO_x limit from the CD.
- Revised the citation in Condition 10.10.2 (lb/ton clinker NO_x limit) to cite Regulation No. 23, as well as the construction permit.

Note that even though the RH NO_x emission limit is the same as the construction permit limit, the RH limit was not streamlined but included in the citation because the RH emission limit cannot be revised without revising the SIP. The construction permit limit was not streamlined since the RH limit is state-only enforceable until approved by EPA into Colorado’s Round 2 Regional Haze SIP. Thus both the CP and RH are cited.

The monitoring language in Colorado Regulation No. 23, Part A, Section V.A.1.b.(ii), except for the sentence regarding hourly clinker production, was streamlined in favor of the language in Condition 10.10.5 (compliance procedure for lb/ton clinker NO_x limit).

- Included the RH 30-day lb/hr NO_x limit as “new” Condition 10.10.3. Conditions that follow are re-numbered. Note that this RH limit was previously included in Condition 10.22.1.1.
- Made some minor revisions to the language in Condition 10.10.3 (monitoring compliance with the annual (ton/yr) NO_x limit).
- Included RH monitoring language (parts of Condition 10.22.3.1 and relevant language from Condition 10.22.3.2) as “new” Condition 10.10.4.
- Removed Condition 10.10.4 (monitoring compliance with the annual (ton/yr) NO_x limit after the RH compliance deadline in Condition 10.22.1) because it is no longer necessary.
- Revised the language in Condition 10.10.5 (monitoring compliance with the lb/ton clinker NO_x limit) to make corrections and clarify requirements and moved it to “new” Condition 10.10.4.3.
- Revised the language in Condition 10.10.6 to reflect the different definitions of operating days for the lb/ton clinker and lb/hr NO_x limits.
- The following changes were made to the SO₂ requirements in Condition 10.12:
 - Revised Condition 10.12 to include the annual (tons/yr) SO₂ RH limit. The annual (tons/yr) SO₂ limit from Construction Permit 12BO444-2 was streamlined (included in Section III.3 of the permit) since the RH limit is more stringent. Condition 10.12 was renumbered as Condition 10.12.1.

- The lb/hr RH SO₂ limit was included as “new” Condition 10.12.2.
- Included RH monitoring language as “new” Condition 10.12.3.
- The language in Condition 10.12.1 (calculating NO_x mass emissions) was revised) and Conditions 10.12.1.1 and 10.12.1.2 were removed.

Condition 10.12.1.1 addressed using hourly emissions for the SO₂ limit in Condition 10.13. Since Condition 10.13 was revised (as discussed later), it was no longer necessary. The language in Condition 10.12.1.2 is replaced by “new” Condition 10.12.3.2.

- Removed Condition 10.12.2. (monitoring compliance with the annual (ton/yr) SO₂ limit after the RH compliance deadline in Condition 10.22.1) because it is no longer necessary.
- The language in Condition 10.13 was revised to provide more specific information on assessing compliance with the facility wide SO₂ limit.

This facility wide limit is 7 lb SO₂ per ton of material processed (including fuel). For purposes of assessing compliance with the limit, tons of material processed will be based on tons of feed to the kiln plus coal burned in the kiln and dryer (if applicable). Since any material processed through the dry would also be processed through the kiln, only feed material to the kiln will be included in the daily tons of material processed.

- Updated the paragraph in Conditions 10.14.1 and 10.16.1 that starts with the phrase “a stack testing protocol shall” to reflect current language regarding test protocol, scheduling and report submittal.
- The following changes were made to Condition 10.16:
 - Condition 10.6.1 was revised to require that the lead performance tests be conducted concurrently with the PM performance tests required by Condition 10.5.1.
 - The note in Condition 10.6.1 was revised to indicate the date of the most recent lead performance test.
 - The paragraph in Condition 10.16.1 that begins with “testing shall be performed for each proposed fuel type” was revised to replace “VOC” with “lead.”
 - Updated the emission factors in the table in Condition 10.16.2 to include the results from the most recent performance tests.
 - Included the CAM requirements as “new” Condition 10.16.4.
- Revised Condition 10.17 (VOC RACT requirement) to update the citation.

Condition 10.17 includes the general RACT provisions in Regulation No. 7, for new or modified sources (this condition simply cites Section II.C – which addresses both new and existing sources). It should be noted that the requirement to conduct RACT

for emissions from new or modified minor sources is also included in Regulation No. 3, Part B, Section III.D. The VOC RACT requirement was not included in the Construction Permit (12BO444-2) but was noted as a requirement in the TRD (see page 17) for the initial permit (issued February 1, 2000) that applied and as a result was included in the initial Title V permit. As previously discussed, Regulation No. 7 was revised in December 2019 and reorganized into parts (e.g. Part A) and then again in April 2023 to split it into separate regulations. As discussed previously for the dryer, the citation will be revised to indicate Regulation No. 24, Part A, Section II.C since that language includes utilizing RACT for both “new” and “existing” sources and RACT was determined to be the design of the kiln (based a modification made to the kiln in 1979-1980).

- The citations in Conditions 10.20 and 10.21 were revised to replace the words “as provided for” with “Construction Permit 12BO444-2, as modified under”. The underlying construction permit, as well as the Title V permit were revised to incorporate the SNCR operating requirements and prohibitions on netting from the CD.
- The RH requirements in Condition 10.22 were moved. The SO₂ and NO_x emission limits and monitoring requirements were moved to Conditions 10.10 and 10.12. The opacity limit and monitoring requirements were included in Condition Note that the annual (ton/yr) emission limitations in Conditions 10.10 and 10.12 were streamlined in favor of the RH annual (ton/yr) emission limitations since the RH annual (ton/yr) limits are more stringent. The requirements in Condition 10.22.2 (compliance date) have passed and are no longer necessary.

Section II.11 – P009, P010, P015, P011, P012 and P013 – Various Material Handling Sources

Note that with the removal of Section II.2, this section is renumbered as Section II.10. The below discussion is based on the numbering in the current permit (last revised November 17, 2017).

- Revised Condition 11.4.3 to include the dates of the most recent performance tests.
- Revised the second paragraph of Condition 11.4.3 to require that the source specifically request in the protocol submittal to test a baghouse that has already been tested and to receive Division approval in writing.
- Revised Condition 11.6.1.7 to indicate that storage piles are temporary and that clinker shall be reclaimed within three (3) days of occurrence. Clinkers is intended to be stored in open piles but in an enclosed space. Clean-up of temporary clinker storage piles is required to be completed within three days.

Section II.13 – P007A – Handling and Processing of CKD and Raw Material Waste Dust

Note that with the removal of Section II.2, this section is renumbered as Section II.12. The below discussion is based on the numbering in the current permit (last revised November 17, 2017).

- Revised Condition 13.2.4 to indicate the date of the most recent performance test.

Section II.14 – General Fugitive Particulate Emissions Requirements

Note that with the removal of Section II.2, this section is renumbered as Section II.13. The below discussion is based on the numbering in the current permit (last revised November 17, 2017).

- Revised the language in Condition 14.1 to specify the records of vehicle miles traveled be maintained.
- Revised Condition 14.2 to add additional information and to specify that records be maintained and made available to the Division upon request (rather than kept on site).
- Revised Condition 14.1.4.3 to require that an employee be stationed at the truck wash when it is not in service to remove cement spillage when the truck wash is not operational.
- Added “new” Condition 14.5 to include the PC MACT requirements for conveying system transfer points. As discussed previously, the description of “process fugitives” in the table in Section I, Condition 4.1 includes “various transfers not vented through a stack (e.g. belt and screw conveyors)”.

Section II.15 – Gasoline Storage Tank

Note that with the removal of Section II.2, this section is renumbered as Section II.14. The below discussion is based on the numbering in the current permit (last revised November 17, 2017).

- The Regulation No. 7 language in this section was revised to reflect minor language changes in the regulation regarding the requirements in Sections IV and XV for gasoline dispensing facilities and to reflect formatting changes to Regulation No. 7. As discussed previously, Reg 7 was revised in December 2019 and reorganized into parts (e.g. Part A) and then again in April 2023 to split it into separate regulations. Therefore, the citations were revised to note that these requirements are now in Regulation No. 24.
- Revisions were made to more appropriately address the applicable requirements in Regulation No. 7 (now Regulation No. 24).

Section II.16 – Daily Visible Emissions Observations

Note that with the removal of Section II.2, this section is renumbered as Section II.15. The below discussion is based on the numbering in the current permit (last revised November 17, 2017).

- Revised the language in Conditions 16.1.1 and 16.1.2 to specify the actions apply if visible emissions are observed during the daily walk through or any other period of operations. While the permit only requires a daily walk through to check for visible emissions, appropriate actions must be taken at any time that visible emissions from a stack or fugitive source are seen.
- Revised Condition 16.1.1.1 to replace “as soon as practicable” with “within one (1) hour”. This language is consistent with the language used in another Colorado Portland Cement plant Title V permit.

Section II.17 – Cold Cleaner Solvent Vats

Note that with the removal of Section II.2, this section is renumbered as Section II.16. The below discussion is based on the numbering in the current permit (last revised November 17, 2017).

- Revised the title for this section to “Solvent Usage.” Added an additional summary table for other solvent cleaning activities and revised the note under the table for cold cleaner solvent vats.
- The Regulation No. 7 language in this section was revised to update the citations. As discussed previously, Reg 7 was revised in December 2019 and reorganized into parts (e.g. Part A) and then again in April 2023 to split it into separate regulations. Therefore, the citations were revised to note that these requirements are now in Regulation No. 25.
- Included more specific language for the solvent cold cleaners.
- As discussed previously, language was added to specify that additional requirements apply to solvent usage if actual, uncontrolled emissions exceed specific levels.

Section II.21 – Particulate Matter Performance Testing

Note that with the removal of Section II.2, this section is renumbered as Section II.20. The below discussion is based on the numbering in the current permit (last revised November 17, 2017).

- Updated the second paragraph to reflect current language regarding test protocol, scheduling and report submittal.

Section II.22 – NESHAP LLL (PC MACT) Requirements

Note that with the removal of Section II.2, this section is renumbered as Section II.21. The below discussion is based on the numbering in the current permit (last revised November 17, 2017).

As discussed previously, revisions were made to the PC MACT requirements since the previous permit issuance. The following changes were made to the permit to address those revisions:

- Change the version date in Condition 22 and removed the paragraph regarding a July 25, 2016 direct final rule.
- The following permit conditions were revised to address revisions made to the PC MACT:
 - Section 63.1348 – compliance requirements: Conditions 22.2.6 (63.1348(a)), 22.12.4 (63.1348(a)(4)(ii)) and 22.13.4 (63.1348(b)(4))
 - Section 63.1349 – performance test requirements: Conditions 22.17.6 (63.1349(b)(1)(vi)), 22.19.4(63.1349(b)(3)(iv) and 22.20 (63.1349(b)(4)(i)).
 - Section 63.1350 – monitoring requirements: Conditions 22.34 (63.1350(g)), 22.34.4 (63.1350(g)(4)), 22.36 (63.1350(j)), 22.38.1 (63.1350(l)(3)), 22.38.1.1 (63.1350(l)(3)(i))and 22.28.1.2 (63.1350(l)(3)(ii))
 - Section 63.1354 – reporting requirements: Conditions 22.52 (63.1354(b)) and 22.53 (63.1354(c))
 - Section 63.1355 – recordkeeping requirements: Condition 22.57 (63.1355(e))
- Removed Condition 22.26 (63.1349(d))

Section II.23 – Compliance Assurance Monitoring (CAM) Requirements

Note that with the removal of Section II.2, this section is renumbered as Section II.22. The below discussion is based on the numbering in the current permit (last revised November 17, 2017).

- Revised Condition 23.1 to reference the CAM plan and to revises Condition 23.1 to be more consistent with the language in the CAM plan and to include the PM CPMS operating parameter.

Section II.25 – Kiln Control Device Support Equipment

Note that with the removal of Section II.2, this section is renumbered as Section II.24. The below discussion is based on the numbering in the current permit (last revised November 17, 2017).

- Deleted the monthly emission and throughput limits from Conditions 25.1 and 25.2, as well as the language preceded by an asterisk (*) under the summary table. These

requirements apply for the first year of operation. The kiln control equipment began operation on July 1, 2016, has completed its first year of operation and the monthly limits no longer apply.

- Removed the paragraphs in Condition 25.1 and 25.2 regarding monthly emissions. In addition, corrected the language in Condition 25.1 to indicate the monthly hours of operation are used to calculate emissions.
- Condition 25.5 in the summary table was revised to replace “hours of operation” with “baghouse operation and maintenance.”
- Removed Conditions 25.6 (commence construction), 25.7 (startup notice) and 25.8 (compliance certification) because these are one time requirements that have been completed.

Section II.26 – Internal Combustion Engines

Note that with the removal of Section II.2, this section is renumbered as Section II.25. The below discussion is based on the numbering in the current permit (last revised November 17, 2017).

- Included the new engine in the summary table titles. Additional revisions to the summary table titles and throughout various text conditions were made to list engine id numbers.
- Requirements for the new engine were included as discussed below and in Section V of this document.
- Since the permit was issued revisions were made to MACT ZZZZ on November 19, 2020 (“once-in-always-in” policy), December 4, 2020 (fuel streamlining), August 10, 2022 (to address the 2015 vacatur) and March 29, 2023 (testing requirements). The language in Condition 26.1 (MACT ZZZ requirements) was revised as follows to address those revisions:
 - Revised the version date.
 - Removed the paragraphs under Condition 26.1 regarding the vacatur and potential revisions to address the vacatur.
 - Added a paragraph under Condition 26.1 noting the proposed revisions were published and that the requirements may change in the future.
 - Revised the language in Conditions 26.1.12 and 26.1.12.2 and removed Conditions 26.1.12.2.b and c.
 - Combined and revised Conditions 26.1.12.3 and 26.1.12.4.
 - Revised Condition 26.1.14.

- Condition 26.3 was revised to specify that hours of operation would be used to calculate emissions for all engines for purposes of APEN reporting requirements. In addition, added the hours of operation that would trigger an APEN submittal for the A-pit pump and the new engine.
- Condition 26.4 was revised to require annual NO_x and CO emission calculations for all engines and include emission factors for all engines. In addition, the paragraph regarding cancelling the APEN for the A-Pit Pump (EGEN-LYO-1) was removed, since the APEN was cancelled. A paragraph was added to address APEN reporting if the APEN de minimis level is exceeded.
- Since the permit was issued revisions were made to NSPS IIII on December 4, 2020 (fuel streamlining), June 29, 2021 (heavy duty engine and vehicle test procedure), August 10, 2022 (to address the 2015 vacatur) and January 24, 2023 (heavy-duty engine and vehicle standards). The language in Condition 26.6 (NSPS IIII requirements) was revised as follows to address those revisions:
 - Revised the version date.
 - Revised the paragraphs under Condition 26.6 regarding the vacatur and potential revisions to address the vacatur.
 - Added a paragraph under Condition 26.6 noting the proposed revisions were published and that the requirements may change in the future.
 - Revised the second paragraph in Condition 26.6.1 (60.4202(a)(2))
 - Revised Condition 26.6.3 to replace “80.510(b)” with “1090.305”. In addition, removed the reference to sulfur standards for LM diesel fuel as this information is not in 1090.305.
 - Revised the language in Condition 26.6.6.3 to update CFR references.
 - Revised the language in Condition 26.6.8 and 26.6.8.2 and removed Conditions 26.6.8.2.b and c.
 - Revised the language in Condition 26.6.8.3.
- Condition 26.7 was revised to include a version date, note that proposed revisions have been published and to address the new trash pump engine (10.7 hp) which is subject to requirements in NSPS Subpart JJJJ.
- Added the NSPS JJJJ requirements to “new” Condition 26.8. These requirements apply to the new trash pump engine (10.7 hp).

Section III – Permit Shield

- The following revisions were made to the permit shield for streamlined conditions (Section III.3):

- Section II, Conditions 5.6 and 5.11 were included in the column labeled “permit condition(s) for the Regulation No. 6, Part B, Sections III.C.2 (PM limit) and III.C.3 (20% opacity requirement).
- Section II, Condition 10.9.3 was included in the column labeled “permit condition(s) for the Regulation No. 6, Part B, Section III.C.3 (20% opacity requirement).
- Added Regulation No. 7, Part E, Section III.A.2 (comply with THC requirements in 40 CFR Part 63 Subpart LLL). The Division interprets this Reg 7 requirement to apply to the kiln and dryer which are subject to THC (or alternative o HAP) requirements in 40 CFR Part 63 Subpart LLL. Note that the citation was updated to Regulation No. 26, Part B, Section III.A.2 (as discussed previously, Regulation No. 7 was split into additional regulations in April 2023).
- Added Regulation No. 23, Part A, Section V.B.2.b (dryer monitoring requirements) to the table. This paragraph (previously Reg 3, Part F, Section VII.C.2.b) was not addressed in the current permit but could have been streamlined.
- Added the NO_x and SO₂ annual (tons/yr) emission limits from Colorado Construction Permits 12BO444-1 (dryer) and 12BO444-2 (kiln).

Section IV – General Conditions

- Changed the version date.
- The following changes were made to Condition 3: minor language changes were made to paragraph 3.d.(ix), the first two paragraphs after 3.d.(x), paragraph 3.g.(viii), and the first two paragraphs after 3.g.(viii).
- Revised General Condition 8 to include potential future fees for major stationary sources located in the 2008 ozone NAAQS nonattainment area.
- Revised General Condition 12 to include requirements in Reg 3, Part D.
- Corrected the citation in Condition 18 (changed from “CCR 1001-17” to “CCR 1001-19”).
- The following changes were made to Condition 22:
 - Revised the language in paragraph e to indicate that an APEN shall be filed once per year for control equipment changes at condensate storage tanks subject to Reg 7, Part B, Section I (previously Reg 7 Section XII and Reg 7, Part D, Section I).
 - Added a paragraph e.(i) to include the new certification requirements in Reg 3, Part A, Section II.A.3. These requirements apply to sources located in the ozone nonattainment area with the potential-to-emit greater than 25 tons/yr of NO_x and/or VOC.

- Added a paragraph f to include the new GHG APEN reporting requirements in Reg 3, Part A, Section II.A.2. These requirements are **state-only** enforceable.
- Added major stationary source monitoring, recordkeeping and reporting requirements in “new” Condition 24. Conditions that follow are renumbered.
- Revised Condition 30 to reflect the April 2023 rulemaking that split Reg 7 into four separate regulations (correct citations and references to include Regulation No. 24) and to note in the introductory paragraph that portions are state-only in certain areas (new nonattainment areas for either the 1-hr or 8-hr ozone standard).

Appendices

- The following changes were made to the insignificant activity list in Appendix A:
 - Removed the “two (2) portable, self-contained conveyors”.
 - Completed the “Note” for the sandstone quarry regarding AIRS pt 052.
- The following changes were made to the tables in Appendices B and C:
 - Revised the engine descriptions to include the source’s identification numbers.
 - Added the 4” trash pump (10.7 hp) engine. The Division’s 2019 inspection report noted that this engine was not included in the permit.
- Revised Appendix D to add a version date, update information on submitting reports to the Division and correct EPA information.
- Cleared the information from the table in Appendix F.
- Revised Appendix G to include a CAM plan for the kiln. As discussed previously, the CAM plan for the kiln was removed in the second renewal (issued March 1, 2017) since CAM would rely on monitoring in the PC MACT. After further review, the Division considers that the CAM plan should not have been removed but should have been revised to rely on the PC MACT monitoring, thus this correction is being made with the third renewal.
- Removed the applicability tests in Appendix I. The projects were conducted in 2015 - 2016 and the source was required to monitor emissions from equipment associated with the project for five years following completion of the project. The five year period ended in 2020 - 2021, therefore, the applicability tests no longer need to be included in the appendix.

Appendix I remains in the event of future projects for which an applicability test would need to be included in the permit. Additional information was included to more thoroughly explain the requirements for projects and the title of this appendix was revised.

VIII. Changes Made to the Draft Permit Based on Comments Received During the Public Comment Period

The modifications in this section were made based on comments that were received during the public comment period and/or errors found in reviewing and addressing public comments. The official public comment period began on October 12, 2023 and ran through November 11, 2023. A Public Comment Hearing was requested during the public comment period and the Hearing was held on January 5, 2024. The public comment period ended at the conclusion of the Hearing.

Note that this does not include administrative-type corrections such as typographical errors that the Division found and made following the release of the draft permit for public comment.

In this Section, the discussion related to modifications is with respect to the draft revised permit that went through public comment. In addition, please be aware that unless otherwise specified changes noted in Sections VI (Modification Requested by the Source) and VII (Other Modifications) were not changed, as a result there may be some inconsistencies between the permit changes noted in those sections and the changes noted in this section.

Section I – General Activities and Summary

- Listed the 2000, 2013 and 2024 Compliance Orders on Consent (COCs) in Condition 1.3.
- The following changes were made to Condition 1.4:
 - Removed Section II, Condition 2.5 from the list of state-only enforceable conditions.
 - Removed the statement that Section IV, Conditions 3.d and 3.g become state-only enforceable on June 1, 2024 and added 3.d to the list of state-only enforceable conditions.
 - Removed the phrase “(last paragraph)” after Condition 3.g, since the entire condition is now state-only enforceable.
- Revised the summary table in Condition 4.1 to indicate the pollution control device for 041 (pug mill mixing, pelletization and truck loading) is wet material rather than a baghouse.

Section II.1 – P017 – Dowe Flats and Lyons Quarry – Fugitive Dust Source

- Moved the paragraph in Condition 1.5.1 regarding weekly inspections to “new” Condition 1.5.5 to increase clarity. In addition, revisions were made to the inspection language and a citation was added (Regulation No. 3, Part C, Section V.C.5.b - periodic monitoring).

- Condition 1.5.1.5 was revised as required by the April 23, 2024 Compliance Order on Consent (COC).
- Included the language in Regulation No. 1, Section III.D.1.d.(i) as “new” Condition 1.5.3.1, conditions that follow are renumbered.
- Reformatted Condition 1.5.4 and corrected the citations.
- Added “new” Condition 1.5.6 to include semi-annual training as required by the April 23, 2024 COC.

Section II.2 - Raw Material Storage and Handling at Plant Site

- Revised the language in Condition 2.1 to indicate that compliance with the daily throughput limit shall be monitored by comparing daily materials processed with the daily throughput limit.
- Revised Condition 2.2 to include appropriate emission factors to calculate PM and PM₁₀ emissions.
- Revised Condition 2.3.2 to require the source to calculate the average monthly truck weight (both empty and full) and capacity (full minus empty weight) from all truck deliveries for the month and to use the number of trucks delivered per day and the monthly average truck capacity to determine daily throughput.
- Included “new” Condition 2.3.3 to require that the monthly average wind speed be monitored and recorded for use in emission calculations.
- Removed the opacity requirements in Condition 2.4. As discussed on page 15, the Division determined that these sources are fugitive emission sources and the Reg 1 opacity limits do not apply.
- Included a requirement to operate and maintain truck weigh scales in accordance with the manufacturer’s recommendations and good engineering practices in “new” Condition 2.4. This condition requires calibration of the scales in accordance with manufacturer’s procedures, on the schedule recommended by the manufacturer.
- Removed the opacity requirement in Condition 2.5. As discussed on page 18, the Division determined that the activities in this section are not manufacturing processes, thus the requirements in Reg 6, Part B, Section III do not apply.
- Included a requirement to monitor and record vehicle miles traveled by the front end loader in “new” Condition 2.5. Frequency of monitoring and recording will be monthly.

January 2025 Revision: The permit that was sent to EPA for their 45-day review period required an odometer to be used to determine vehicle miles traveled (VMT).

The permit was revised to allow use of either an odometer or global positioning system (GPS) prior to permit issuance. The Division considers that this is a correction and meets the requirements for an administrative amendment in Reg 3, Part A, Section I.B.1. In correspondence with the source between the public comment and EPA review periods, the Division indicated that CEMEX use some device (odometer or GPS) to measure VMT for the front end loader, a GPS was included in the Division's proposed permit language. CEMEX indicated that they could use an odometer but after the proposed permit went to EPA for review, CEMEX indicated the front end loader was not equipped with an odometer. As a result, the permit was revised to allow either the odometer or a GPS. Both devices measure distance traveled and were contemplated for use in "new" Condition 2.5. The Division considers that the GPS is more accurate than an odometer and considers the change an improvement over the language that went to EPA for review.

- Removed Condition 2.6.1.1 (height of discharge from the stacker belt shall be adjusted to minimize the drop height...). The stacker belt was associated with the conveyor that brought raw materials from Dowe Flats Quarry to the plant. Since the Dowe Flats conveyor is no longer used and the permit for the conveyor is cancelled, the stacker belt is no longer used and is in the process of being removed. Therefore this control measure is no longer relevant and was removed.
- Removed Condition 2.6.1.2 (transfer points shall be enclosed and maintained under negative pressure). This control measure was included in the underlying construction permit (98BO0292) when it was first issued on September 11, 1998. Construction permit 98BO0292 was issued to allow for construction for an underground reclamation system for the patio storage and handling area. The underground reclamation system was not installed and construction permit 98BO0292 was modified on August 17, 2000 to reflect that. The control measure in Condition 2.6.1.2 (transfer points shall be enclosed and maintained under negative pressure) was removed from Construction Permit 98BO0292, unfortunately that revision was not reflected in the Title V permit. Therefore, the Division is correcting this issue now.
- Moved the paragraph in Condition 2.6.1 regarding weekly inspections to "new" Condition 2.6.5 to increase clarity. In addition, revisions were made to the inspection language and a citation was added (Regulation No. 3, Part C, Section V.C.5.b - periodic monitoring).
- Condition 2.6.1.7 was revised as required by the April 23, 2024 Compliance Order on Consent (COC).
- Added "new" Condition 2.6.6 to include semi-annual training as required by the April 23, 2024 COC.

Section II.4 – Raw Materials Drying

- Condition 4.2 was revised to require monitoring of dryer feed material daily.
- Condition 4.3 was revised to more specifically identify the requirements related to coal consumption and to require that coal use be monitored and recorded daily.
- Minor revisions to Condition 4.4 were made to specify that natural gas consumption be recorded monthly and coal consumption be monitored in accordance with Condition 4.3.
- The following changes were made to Condition 4.6:
 - Revised the last sentence in Condition 4.6 to state “[c]ompliance with the PM, PM₁₀ and lead emission limits shall be monitored as follows:”
 - Included requirements for a bag leak detector required by previous COCs to Condition 4.6.1.
 - Revised the last sentence in the last paragraph of Condition 4.6.2 to replace “daily” with “hourly”. In addition, revised Condition 4.6.2 to include stack test requirements for lead emissions and to note the date the most recent lead test was conducted.
 - Removed the phrase “[i]n addition to the stack tests described above (Condition 4.6.2)” from the beginning of Conditions 4.6.3 and 4.6.4 because it is not necessary.
 - Revised Condition 4.6.3 to include the lead emission factors and included the appropriate citation for PM₁₀ and lead emissions.
 - Revised Condition 4.6.4 to note that lead emissions are also subject to the compliance assurance monitoring requirements.
 - Added “new” Condition 4.6.5 to require the source to monitor compliance with the short-term (hourly) feed and coal consumption limits.
- The following changes were made to Condition 4.7:
 - Added “new” Condition 4.7.3 to require the source to comply with the fuel restriction requirements.
 - Added “new” Condition 4.7.4 to require the source to monitor comply with the combustion process adjustment requirements.
- The following changes were made to Condition 4.8:
 - Revised Condition 4.8.1 to indicate that performance tests are required for CO and VOC. Removed information related to lead testing and most recent lead test date.
 - Removed the lead emission factors from the table in Condition 4.8.2.

- Added “new” Condition 4.8.3 to require the source to comply with the combustion process adjustment requirements.
- Revised the language in Condition 4.12 to indicate CAM applies with respect to the PM, PM₁₀ and lead emission limits.

Section II.9 – Kiln

- Revised the citation in Conditions 9.11 and 9.14 to include Colorado Regulation No. 3, Part C, Sections I.A.7 and III.B.7.
- Revised Condition 9.16 of the summary table to include baghouse operation and maintenance as a monitoring method for the lead emission limit. The baghouse operation and maintenance requirements were already included as Condition 9.16.3.
- Revised the language in Condition 9.19 to indicate CAM applies with respect to the PM, PM₁₀ and lead emission limits.

Section II.10 – P009, P010, P015, P011, P012 and P013 – Various Material Handling Sources

- Moved the paragraph in Condition 10.6.1 regarding weekly inspections to “new” Condition 10.6.5 to increase clarity. In addition, revisions were made to the inspection language and a citation was added (Regulation No. 3, Part C, Section V.C.5.b - periodic monitoring).
- Condition 10.6.1.9 was revised as required by the April 23, 2024 Compliance Order on Consent (COC).
- Added “new” Condition 10.6.6 to include semi-annual training as required by the April 23, 2024 COC.

Section II.12 – P007A – Handling and Processing of CKD and Raw Material Waste Dust

- Revised Condition 12.2.5 to include “new” condition 12.2.5.4 which requires the source to comply with the fugitive particulate matter control requirements in Condition 12.7.
- Moved the paragraph in Condition 12.7.1 regarding daily inspections to “new” Condition 12.7.5 to increase clarity. In addition, revisions were made to the inspection language and a citation was added (Regulation No. 3, Part C, Section V.C.5.b - periodic monitoring).
- Condition 12.7.1.7 was revised as required by the April 23, 2024 Compliance Order on Consent (COC).

- Added “new” Condition 12.7.6 to include semi-annual training as required by the April 23, 2024 COC.

Section II.13 - General Fugitive Particulate Emissions Requirements

- Revised Condition 13.4 of the summary table under the columns “Monitoring – Method and Interval” to replace “certification” and “semi-annually” with “inspections” and “weekly.”
- Revised Condition 13.4.1.3 to include a citation to indicate where this requirement came from.
- Included “new” Condition 13.4.5 to require weekly inspections of fugitive emission sources (e.g. haul roads, open storage piles) not subject to emission limits.
- Added “new” Condition 13.4.6 to include semi-annual training as required by the April 23, 2024 COC.

Section II.15 – Daily Visible Emission Observations

- This condition was revised as follows:
 - Revised Condition 15.1 to require that daily visible emission observations be conducted when plant equipment is operating.
 - Removed the exceptions to daily visible emissions observations in Condition 15.3. Condition 15.3 was revised to indicate that if plant equipment is not operating at any time during the day, the daily observations may be made at any time during the day.
 - Other minor language changes were made.

Section II.17 - Continuous Emission Monitoring and Continuous Opacity Monitoring Systems

- Revised Condition 17.2.7 to replace the word “alternative” with “item” since this permit condition does not address alternatives.

Section III – Permit Shield

- The following revisions were made to the permit shield for streamlined conditions (Section III.3):
 - Section II, Condition 2.2 was removed from the column labeled “permit condition(s) for the Regulation No. 6, Part B, Sections III.C.2 requirement (PM limit).

Section IV – General Conditions

- Changed the version date.
- Added a sentence to the end of Condition 2.b indicating compliance certifications are to be submitted via email.
- Removed the language in Condition 5 (emergency provisions) and labeled it as “repealed July 15, 2024.”
- Removed the last sentence in Condition 21 and added an email address to paragraph c.
- Revised Condition 22 to add a sentence at the end of paragraph d indicating reports are to be submitted via email and removed “or V” after “Regulation No. 7, Part B” in paragraph f.

Appendices

- A note was added under the category “non-road engines” in the insignificant activity list in Appendix A to indicate that these engines are not stationary sources.
- Revised Appendix B to remove references to “emergency” or “emergency provisions.”

IX. Environmental Justice Considerations

The following information is provided as a general overview for efforts currently underway at the Department and the Division.

1. CDPHE’s Overall Commitment to Environmental Justice

As is the case in many states, Colorado has a legacy of historic policies, such as redlining and zoning laws, that led to racial and income segregation in housing. Low-income communities and communities of color were and are often located in close proximity to industrial areas, highways, and other pollution sources. The shadows of these housing and land use patterns persist today, with low-income communities and communities of color in Colorado continuing to face greater environmental health risks. The physical environment, where people live, work, learn and play, is also a determinant of health. An individual’s environment can facilitate positive health and well-being or it can produce an unhealthy level of exposure to pollution, contaminants, and/or disease.

On July 2, 2021, Governor Polis signed the Environmental Justice Act into law. This act prioritizes reducing environmental health [disparities](#) in [disproportionately impacted communities](#). It also includes climate-related provisions. Specifically, it recognizes that “all people have the right to breathe clean air, drink clean water, participate freely in decisions that affect their environments, live free of dangerous levels of toxic pollution, experience equal protection provided by

environmental policies, and share the benefits of a prosperous and vibrant pollution-free economy.”

A core component of the Environmental Justice Act is the requirement to systematically identify and reduce environmental health risks in disproportionately impacted communities. Correspondingly, CDPHE has established an [Environmental Justice Program](#) to support all of CDPHE, including the Division, in improving meaningful two-way engagement with low-income communities and communities of color. The goal is to ensure that environmental regulations and policy decisions reflect the communities they will affect, and that adverse environmental impacts are reduced through strategic decisions in regulations, policies, and permits. CDPHE is committed to being a national leader in the environmental justice field.

CDPHE has implemented a number of environmental justice focused actions, including:

- **Identifying and Mapping Disproportionately Impacted Communities.** In June 2022, CDPHE launched a new interactive environmental justice mapping tool, [Colorado EnviroScreen](#). The tool shows areas that meet the definition of disproportionately impacted community and more generally identifies areas with increased environmental health burdens and cumulative impacts. Colorado EnviroScreen is intended for use by permitting staff throughout the Department, including within APCD, for conducting screening and analysis to identify areas with potential environmental justice concerns.
- **Establishing an Environmental Justice Advisory Board.** Created by statute, the Advisory Board meets quarterly. It responds to environmental justice policy questions from the Governor’s Office and CDPHE. For example, in 2022, the Advisory Board fielded a policy question on carbon capture, utilization, and storage from the Governor’s Office. It also advises CDPHE on best practices for engagement with disproportionately impacted communities. On November 21, 2022, the Board launched a [new environmental justice grants program](#) to fund projects that avoid, minimize, mitigate, or measure adverse environmental impacts in disproportionately impacted communities, and to increase meaningful engagement of disproportionately impacted communities in rulemaking and permitting proceedings.
- **Creating the Environmental Justice Ombudsperson position.** Governor Polis appointed Marcus Howell as Colorado’s first Environmental Justice Ombudsperson in February 2022. He reports directly to CDPHE Executive Director Jill Hunsaker Ryan and receives administrative support from the Environmental Justice Program, but is otherwise independent. His duties include advocating for and liaising with disproportionately impacted

communities; working for improved relationships with disproportionately impacted communities; and overseeing CDPHE’s response to complaints and inquiries about environmental justice matters. The Ombudsperson launched a [new environmental justice complaint system](#) on January 20, 2023.

- **Completing Environmental Justice Action Task Force recommendations.** After nearly a year of extensive community outreach, the 22-member Environmental Justice Action Task Force submitted its final recommendation to the state legislature, Governor’s office, and CDPHE on November 14, 2022. The task force solicited community feedback from across the state, holding in-person and virtual meetings in English and Spanish, and receiving more than 300 written comments over the year. The task force listened to numerous presentations from community members and leaders. Its six subcommittees met for a combined 77 hours. The full report is available [here](#). Several of the Task Force recommendations involve methods of incorporating environmental equity and cumulative impact analyses (“EECIA”) into permitting decisions. If adopted by the legislature, these recommendations could significantly expand the Division’s authority to incorporate environmental justice considerations into Title V permitting actions.
- **Enhancing Outreach to Disproportionately Impacted Communities.** CDPHE has substantially expanded its outreach and engagement with disproportionately impacted communities about actions and topics such as air quality, water quality, and solid and hazardous waste cleanup. As part of this broader commitment to meaningfully engage disproportionately impacted communities, CDPHE has implemented the Environmental Justice Act’s requirements to engage disproportionately impacted communities in air quality rulemakings. These requirements include specific timelines for notice, ensuring specific methods of engagement, language justice (translation and interpretation) practices, providing plain language fact sheets, and holding multiple outreach sessions at different times of day and days of the week to ensure maximum accessibility. The APCD Climate Change Program, Regulatory Development team, Office of Innovations in Policy, Environmental Justice Program, and CDPHE’s Office of Boards and Commissions have been providing enhanced community engagement opportunities for rulemakings with significant potential to secure benefits and reduce harms to disproportionately impacted communities. The Division has also employed the enhanced outreach practices in the Environmental Justice Act in other capacities even where there is no legal requirement to do so.

2. Environmental Justice in Title V Permitting

A. History, Background, and the Status Quo

A core component of environmental justice is meaningful community involvement. Since its inception, the Division's Title V permitting program has embodied that principle by providing all legally required opportunities for public comment. The Division adheres to all public notice procedures in federal and state statutes and regulations. The Division also accepts public comment on Title V permits for at least the minimum time required by state and federal law. If a member of the public requests a public comment hearing on a Title V permit, the Division works with Boards and Commissions staff to convene a public comment hearing presided over by one or more members of the Air Quality Control Commission. Finally, Division permitting staff provide responses to significant public comments received both in writing and verbally during public comment hearings.

The Division has also long recognized that it is important to go beyond these minimum public notice and comment requirements in some cases. Residents of disproportionately impacted communities face many additional barriers (including time, money, education, and language challenges) that make it more difficult for them to engage in the Division's ordinary notice and comment processes. As a result, in April 2017 CDPHE adopted [Guidance for Incorporating Environmental Justice into the State of Colorado's Environmental Permitting Procedures](#). This guidance included an appendix with specific guidance for the Division. It outlines a process of prescreening, assessment, relevancy planning, enhanced outreach, permit drafting, and evaluation. The guidance applies to initial applications for Title V operating permits.

The Division has adhered to the 2017 guidance since it was adopted, and is currently in the process of updating it. First, the CDPHE (internal-only) Health Equity & Environmental Justice (HE&EJ) Data Viewer, which was used for identifying permits in the screening process, is no longer being maintained and updated. It has been replaced with the external-facing, publicly available Colorado EnviroScreen interactive environmental justice mapping tool. Second, the Environmental Justice Act adopted a new definition of "disproportionately impacted community" that is substantially broader than the narrower criteria used to identify the most impacted communities in the previous internal HE&EJ Data Viewer, warranting further discussion about which areas should be considered in screening analyses. Finally, numerous new requirements and guidance documents have been created or are currently being considered at both the state and federal level around the substantive aspects of how environmental justice should be incorporated into air quality permitting generally, including Title V permitting.

1. Meaningful Involvement

Meaningful involvement of all people regardless of their race, color, national origin, preferred language, or income is a key component of environmental justice. Under the 2017 guidance and the Division's

ongoing practice, for facilities where environmental justice concerns are raised, a key aspect of the Division's response is ensuring meaningful involvement of the impacted community through enhanced outreach practices.

2. Fair Treatment

Environmental justice incorporates not only meaningful involvement of disproportionately impacted communities, but also fair treatment of all people regardless of their race, color, national origin, preferred language, or income. In the context of air quality regulation, ensuring fair treatment generally involves ensuring that certain communities or populations are not exposed to disproportionately high levels of pollution that could present potential health risks.

Because Title V is a program that aggregates all state and federal requirements associated with facility operations that emit air emissions, it is not an emissions authorizing program. Title V permits are a critical tool for the Division's enforcement and compliance staff to use in enforcing statutes, regulations, and permit terms. To aid this goal, many of Colorado's Title V permits include compliance assurance measures, monitoring, recordkeeping, and reporting requirements built into them.

B. Next Steps and Moving Forward

The Division has made significant strides towards advancing environmental justice in its Title V permitting program to date, but recognizes more work is needed. Given the rapidly evolving nature of both state and federal legal requirements and guidance related to environmental justice in air permitting, the Division is working on updates to its 2017 [Guidance for Incorporating Environmental Justice into State of Colorado Environmental Permitting Procedures](#). These forthcoming updates will reflect changes made in light of many recently released documents and new tools including:

- New requirements of and authority provided by the [Environmental Justice Act](#) (HB21-1266) (July 2021);
- U.S. Environmental Protection Agency ("EPA") [Legal Tools to Advance Environmental Justice](#) (May 2022) with [Cumulative Impacts Addendum](#) (January 2023);
- Release of [Colorado EnviroScreen](#) version 1.0 (June 2022);
- [EPA Interim Environmental Justice and Civil Rights in Permitting Frequently Asked Questions](#) (August 2022);
- [Final Recommendations of the Environmental Justice Action Task Force](#) ("EJATF" or "Task Force") (November 2022);
- EPA, [EJ in Air Permitting: Principles for Addressing Environmental Justice Concerns in Air Permitting](#) (December 2022); and

- [Air Quality Control Commission rulemaking on enhanced permitting for facilities located in disproportionately impacted communities](#) (rulemaking hearing scheduled for May 2023).

The updated guidance will outline a three stage process for stationary source permits, including Title V permits, that includes screening for environmental justice concerns, analyzing potential environmental justice concerns and mitigating potential adverse disparate impacts. The Division anticipates completing the updated guidance later in 2023.

The Division has also proposed new rules before the Air Quality Control Commission regarding permitting processes in disproportionately impacted communities. The rules were proposed on January 19, 2023. This rulemaking is required by the Colorado Environmental Justice Act, but the Division has proposed several provisions that go beyond the minimum statutory requirements.

The proposed rules would apply to both new and modified sources seeking construction permits and most Title V operating permit actions. The proposed rules articulate requirements for enhanced modeling and monitoring in disproportionately impacted communities and require permit applicants to include environmental justice efforts in their permit application. The Air Quality Control Commission is scheduled to hold a rulemaking hearing on the proposal in May 2023.

The Division will incorporate the final version of that rulemaking into its updated guidance document. The proposed rules would clarify and expand upon the Division's capacity to systematically build environmental justice into its Title V permitting processes in the future.

Potential to Emit

Process	Plant ID	AIRS Pt No	Description	Basis for PTE	Emissions (tons/yr)							
					PM	PM ₁₀	SO ₂	NO _x	CO	VOC	Pb	HAPs ¹
Dowe Flats/Lyons Quarry - Fugitive Emissions	P017	017	Blasting - combustion by-products	permit limits			1.2	10.0	39.6			
		025	Fugitive PM - Dowe Flats	permit limits	134.2	58.4						
			Fugitive PM - Lyons Quarry, disturbed areas	permit limits	19	9.4						
Dowe Flats Quarry - Pt Source Emissions		026	Conveyor	permit limits (includes both crusher and conveyor)	0.16	0.07						
		027	Primary Crusher									
General Fugitive	P018	028	Process Fugitives - Lyons Plant	2014 actuals x 1.2	47.33	6.54						
		019	Haul Roads - Not Subject to Emission Limitations	2017 actuals x 1.2	12.84	5.76						
Raw Material Storage and Handling at Plant Site	P000	024	Disch of primary crushed mtl/front end loader	permit limits	15.5	7						
Primary Crusher (plant)	P001	001	Primary crushing/surge silo	emission factors in T5 permit and throughput limits on related equip.	0.54	0.54						
Raw Mtls Drying	P002	002	Dryer	permit limits	22.8	22.8	36.7	13.9	57.3	144.8	1.6	
Secondary Crushing	P003	003	secondary crushing/screening, belt trans.	emission factors in T5 permit and throughput limits on related equip.	0.18	0.18						
Raw Mtl Storage Silos	p004	004	raw mtl silos	emission factors in T5 permit and throughput limits on related equip.	0.048	0.048						
Raw Mtl Grinding	P005	005	raw mtl grinding	emission factors in T5 permit and throughput limits on related equip.	33.15	33.15						
Homogenizing and Blending	P006	006	silos	emission factors in T5 permit and throughput limits on related equip.	0.032	0.032						

CEMEX Construction Materials South, LLC – Lyons Cement Plant
Operating Permit No. 95OPBO082
Technical Review Document – Third Renewal

Process	Plant ID	AIRS Pt No	Description	Basis for PTE	Emissions (tons/yr)							
					PM	PM ₁₀	SO ₂	NO _x	CO	VOC	Pb	HAPs ¹
Kiln Burning ²	P007	007	Kiln	For PM and PM ₁₀ - MACT LLL limit (2015) and clinker limits, for SO ₂ and NO _x - RH ton/yr limits and for other pollutants (except HAPs) permit limits.	21	21	95	901	396	138	4.4	11.43
Clinker Cooler	P009	008	Clinker Cooler	MACT LLL limit (1999/ [2015]) and kiln feed limits. Assume PM = PM ₁₀	21	21						
Clinker & Gypsum/Additive Silos and Weigh Feeders	P009	009	feeders, silos, transfer belts	permit limits	9.3	4.65						
Sheltered (A-Frame) Clinker Storage and Reclaim	P010	010	Reclaim Feeder, A-Frame, Transfer belts	permit limits	21.96	10.98						
Outdoor Clinker Storage & Handling	P015	015	Outdoor clinker storage pile	permit limits	2.05	0.92						
Cement Finish Mill and Auxiliaries	P011	011	Finish Mill	permit limits	17.05	8.65						
	P012	031	Finish Mill Separator	permit limits	21.9	10.95						
Cement Silos/Packhouses/ Loadout	P013	013	silos, packhouses	permit limits	12.3	6.2						
Material Handling System - Load-in and Load-out	P014	014	Silo, loader - coal and clinker	emission factors in T5 permit and throughput limits on related equip.	0.051	0.051						
Handling and Processing of CKD & Raw Mtl Waste Dust	P007A	049	dust silos	permit limits	19.95	9.98						
			pug mill/truck loading	permit limits	2.66	2.66						
			hauling and disposal	permit limits	5.5	2.5						
Rail Car Unloading	P050	050	Rail Car Unloader and Handling System	permit limits	0.6	0.4						
Lime Storage Silo	LIS-1	055	Lime Silo	permit limits	0.004	0.004						
Lime Weigh Hopper	LIS -2	054	Lime Weigh Feeder	permit limits	0.004	0.004						
Total					441.11	243.87	132.90	924.90	492.90	282.80	6.00	11.43

¹HAPs from the kiln include only HCl emissions and are based on the emission factor included in the October 23, 1998 revised T5 application and the allowable clinker production limit (600,000 tons/yr)

²The annual (tons/yr) PM and PM₁₀ emission limits from the construction permit (12BO444-2) for the kiln are higher than the allowable limits based on the MACT LLL limit and allowable clinker production rate. However, the more stringent MACT LLL requirements limit PM and PM₁₀ emissions, although the higher annual limits (133 tons/yr) remain in the Title V permit. Note that the construction permit (12BO444-2) included higher annual (tons/yr) SO₂ and NO_x limits, but these limits were streamlined in favor of the lower RH limits.